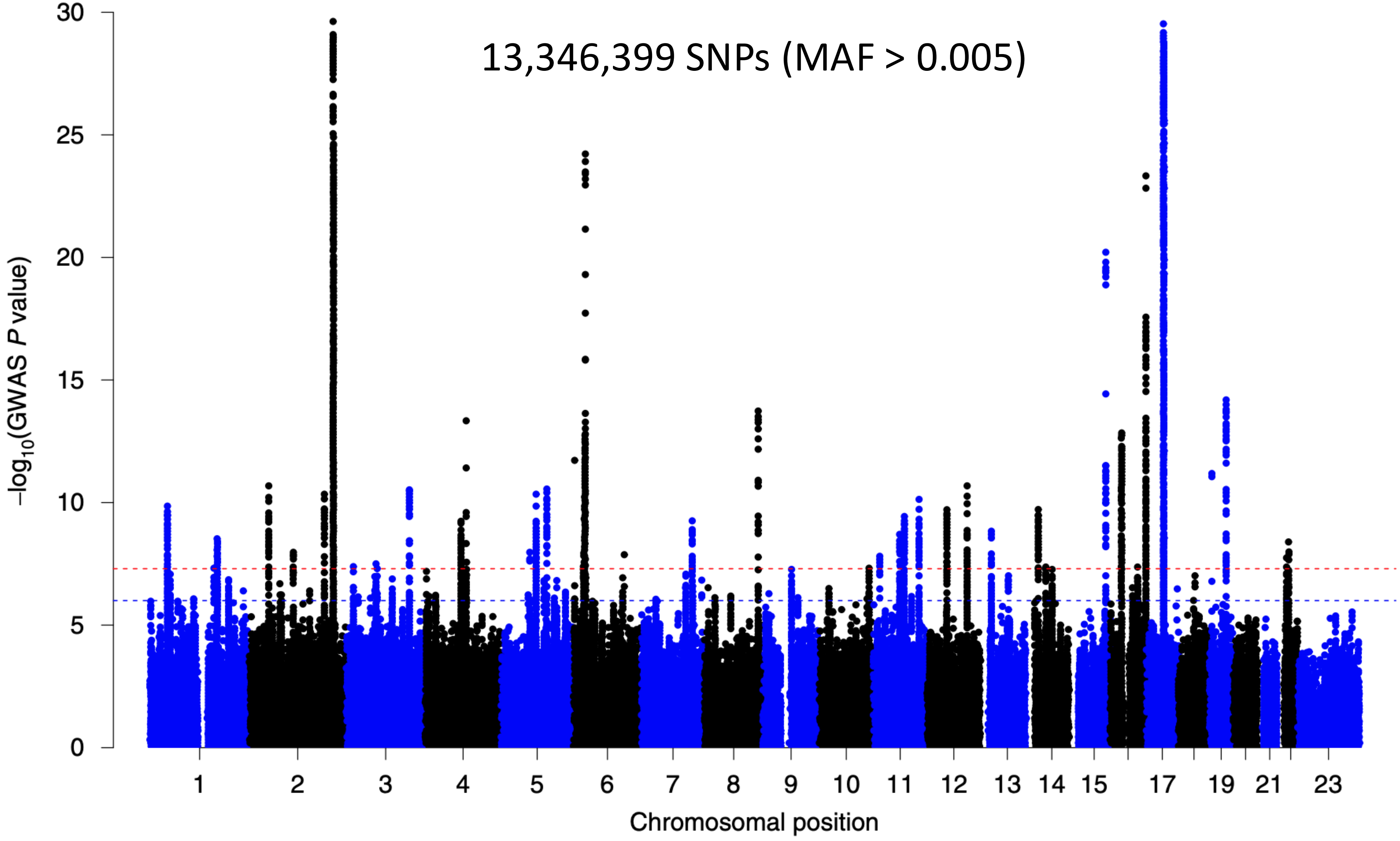
Genomic Visualization Part 1

HUGEN 2073

Genomic Data Visualization & Integration

Slides adapted from Ryan Minster with permission



Cuellar-Partida *et al. Nat Hum Behav* 2021

Learning Objectives

By the end of the session, students will be able to:

- Describe taxonomies for genomic map visualization
- Describe the difference between vector and raster graphics, use each in appropriate circumstances

Taxonomies for Genomic Map Visualizations

Visualization Taxonomy

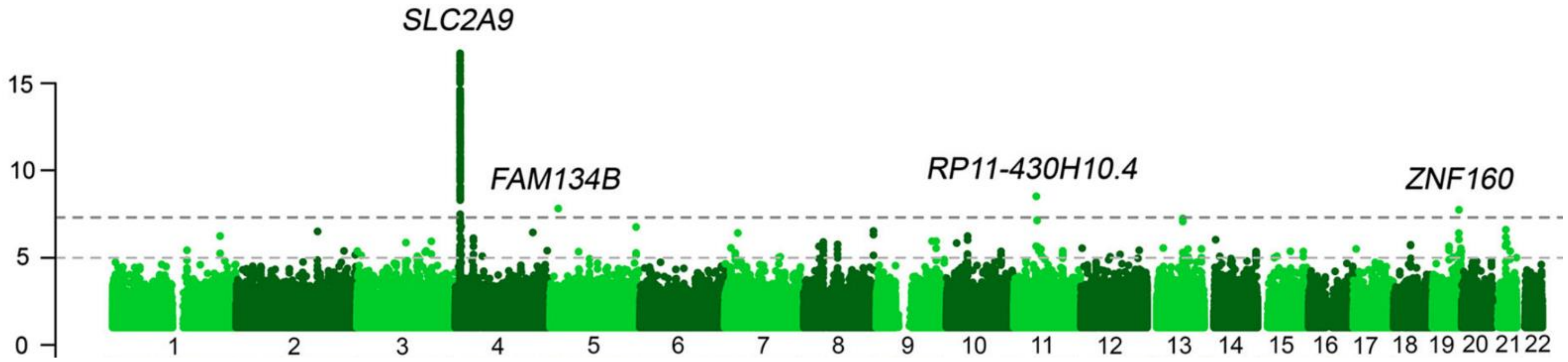
- Layout
- Partition
- Abstraction
- Arrangement

Layout

- Linear
- Circular
- Space-filling
- Spatial

Linear Layouts

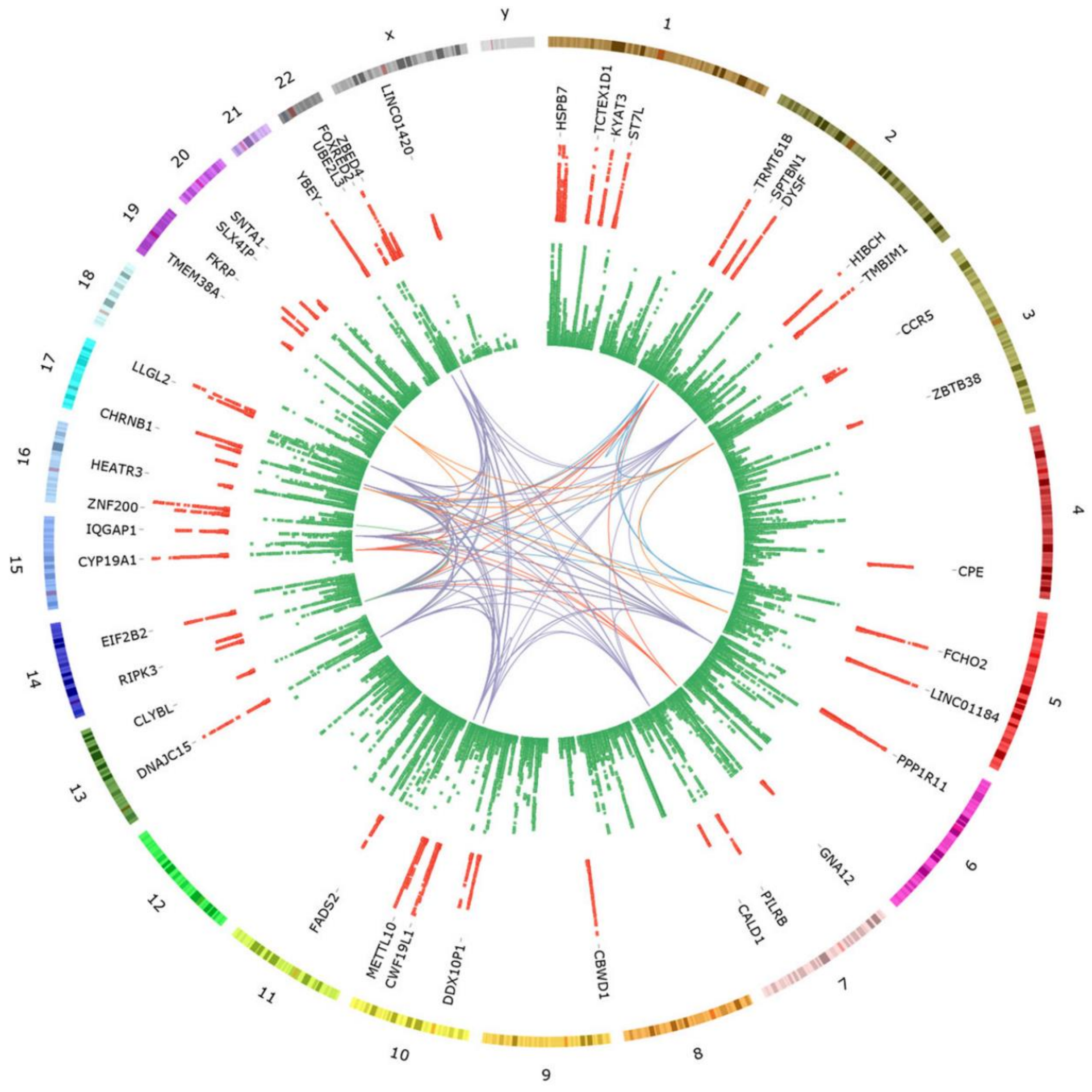
- Manhattan plots
- Most genome browsers (UCSC etc.)



Nagy *et al*. *Genome Med* 2017

Circular Layout

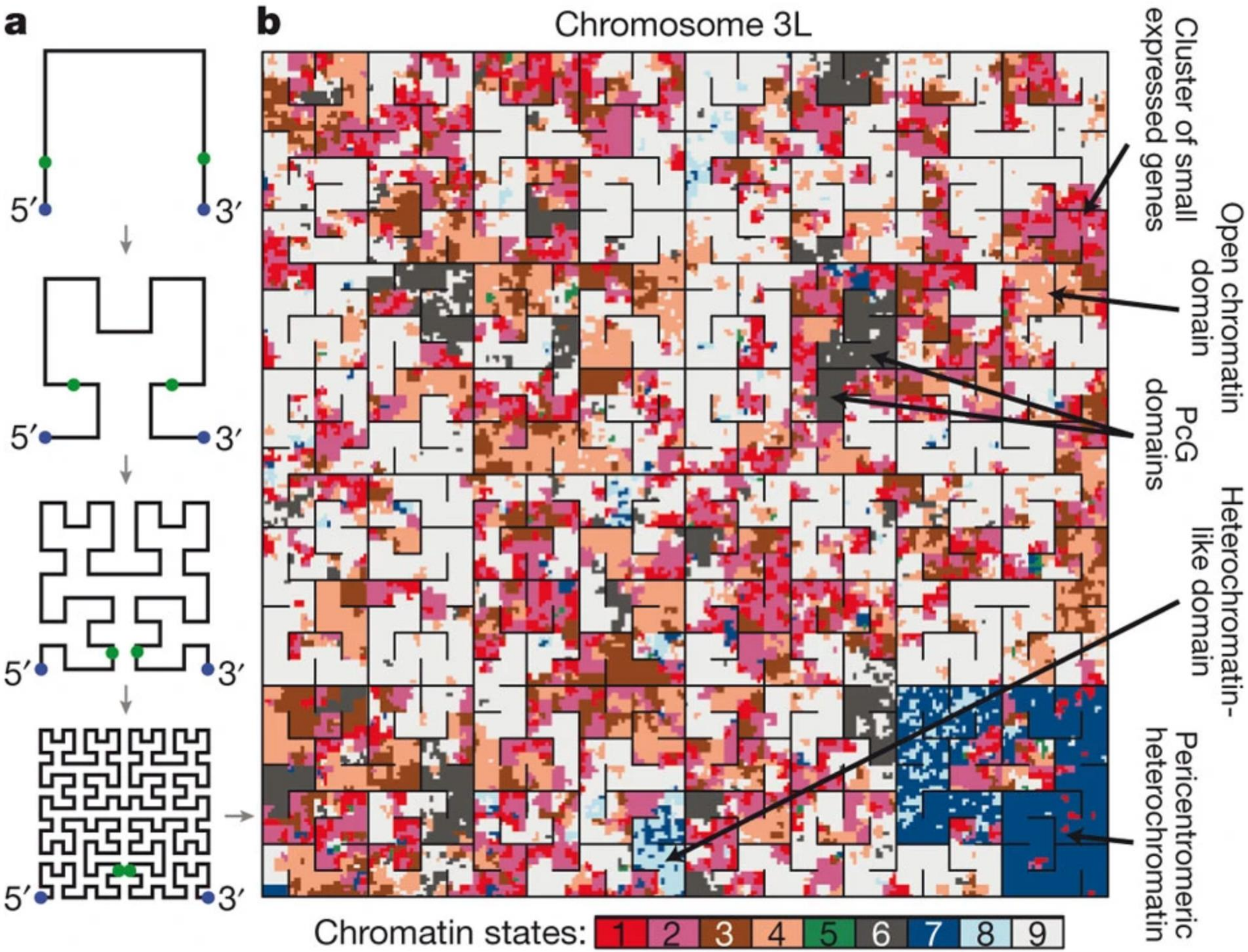
- “Circos” plots



Mullin *et al.* *Genome Biol* 2020

Space-Filling Layout

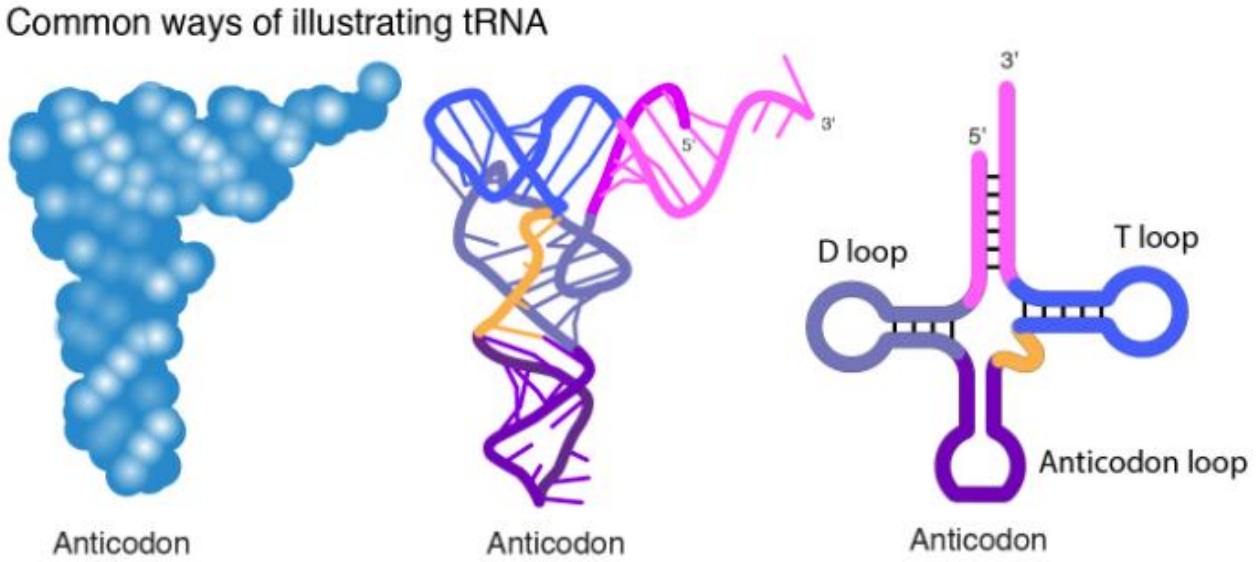
- e.g., Hilbert curves



Kharchenko *et al*. *Nature* 2010

Spacial Layout

- Actual (or hypothetical or abstract) 3D Structure



tRNA. NHGRI Talking Glossary of Genetic Terms

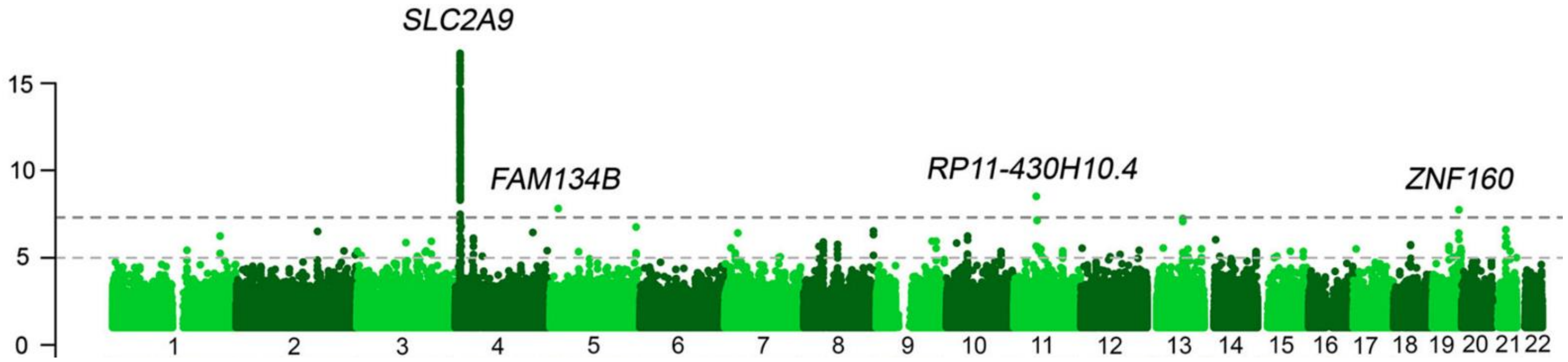
tRNA. Wikipedia

Partition

- Contiguous
- Segregated

Contiguous Partition

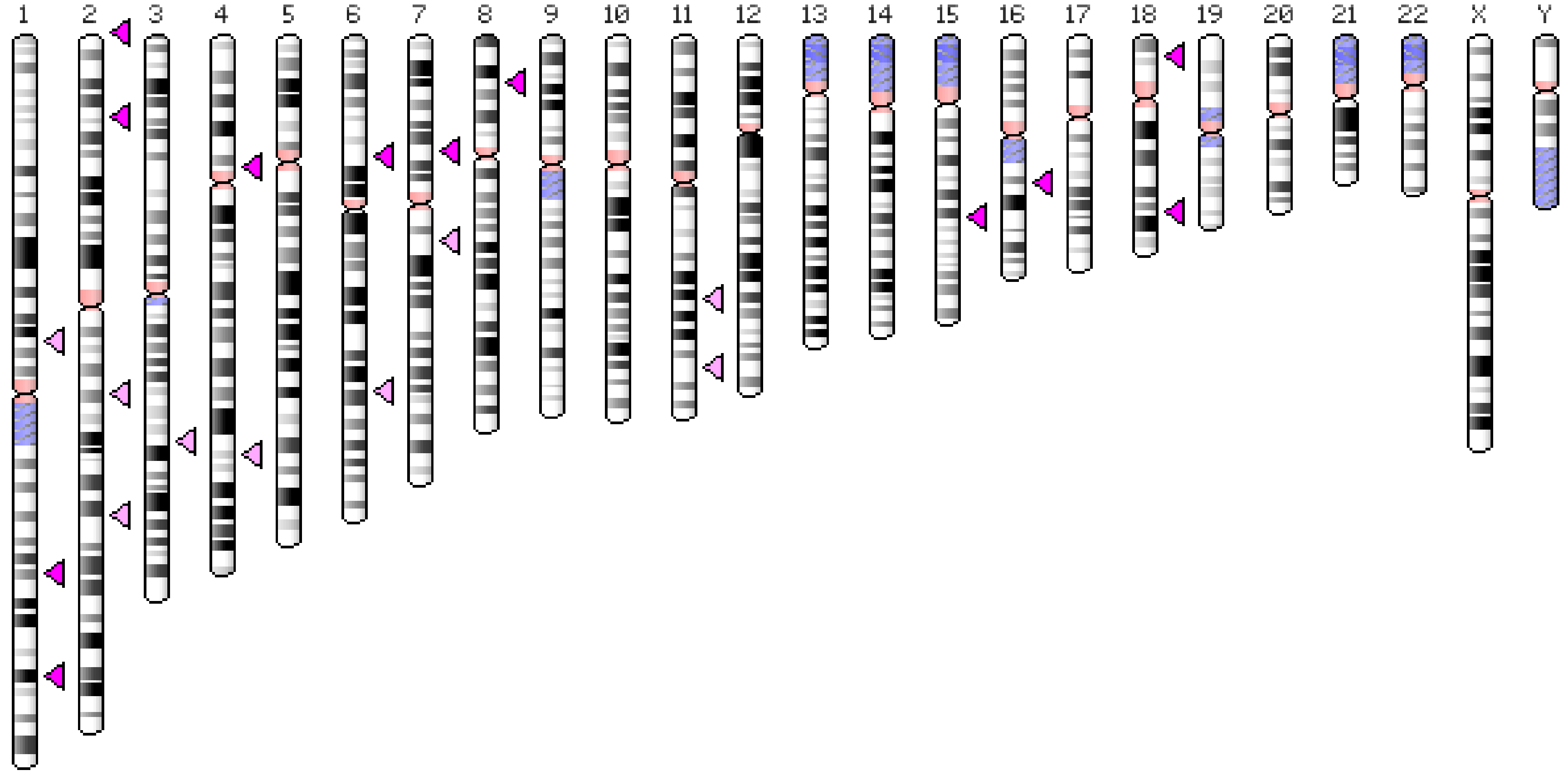
- Manhattan plots
- Genome browsers



Nagy *et al. Genome Medicine* 2017

Segregated Partition

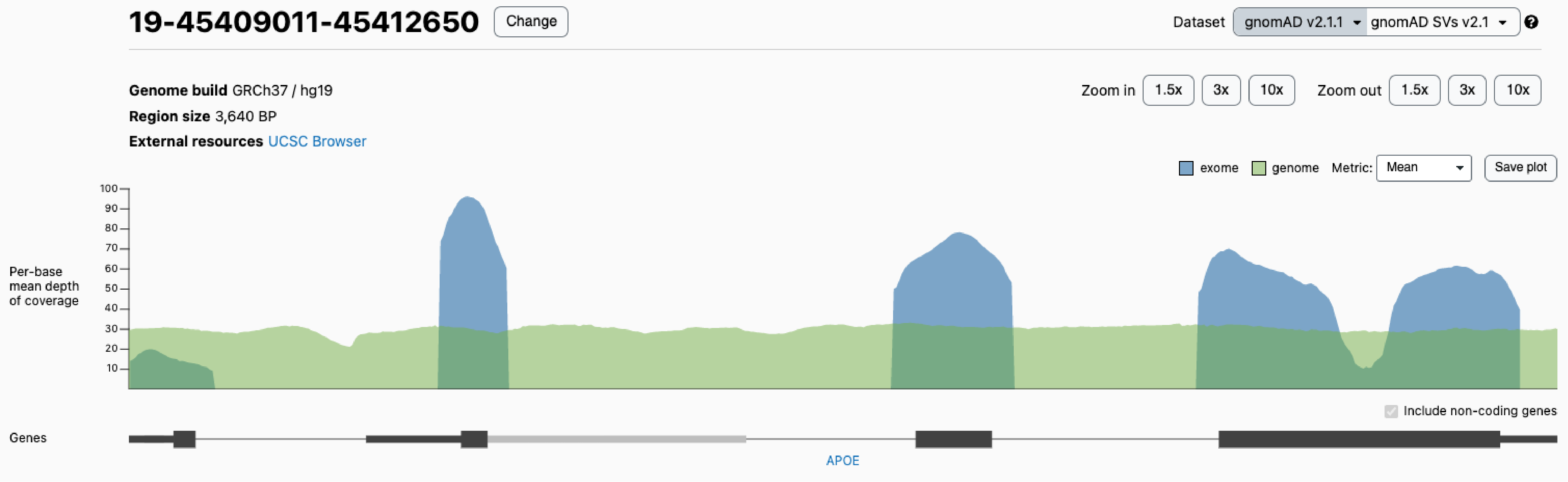
- Ideograms



BMI loci. PheGenI

Abstraction

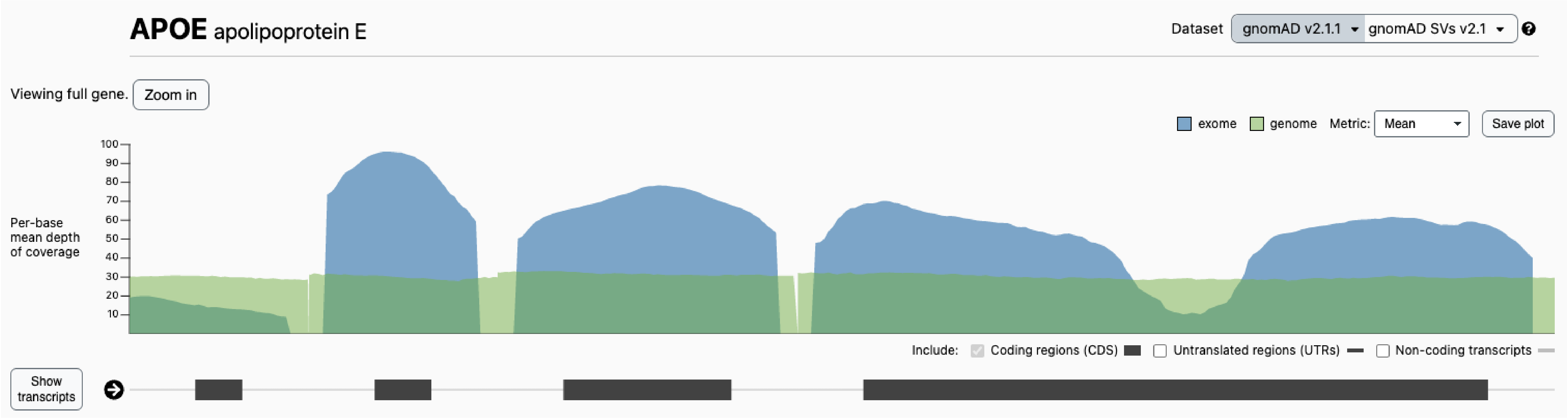
- Replace segments of sequence with abstract symbols



19:45409011–45412650. gnomAD. 2022 Feb 17

Abstraction

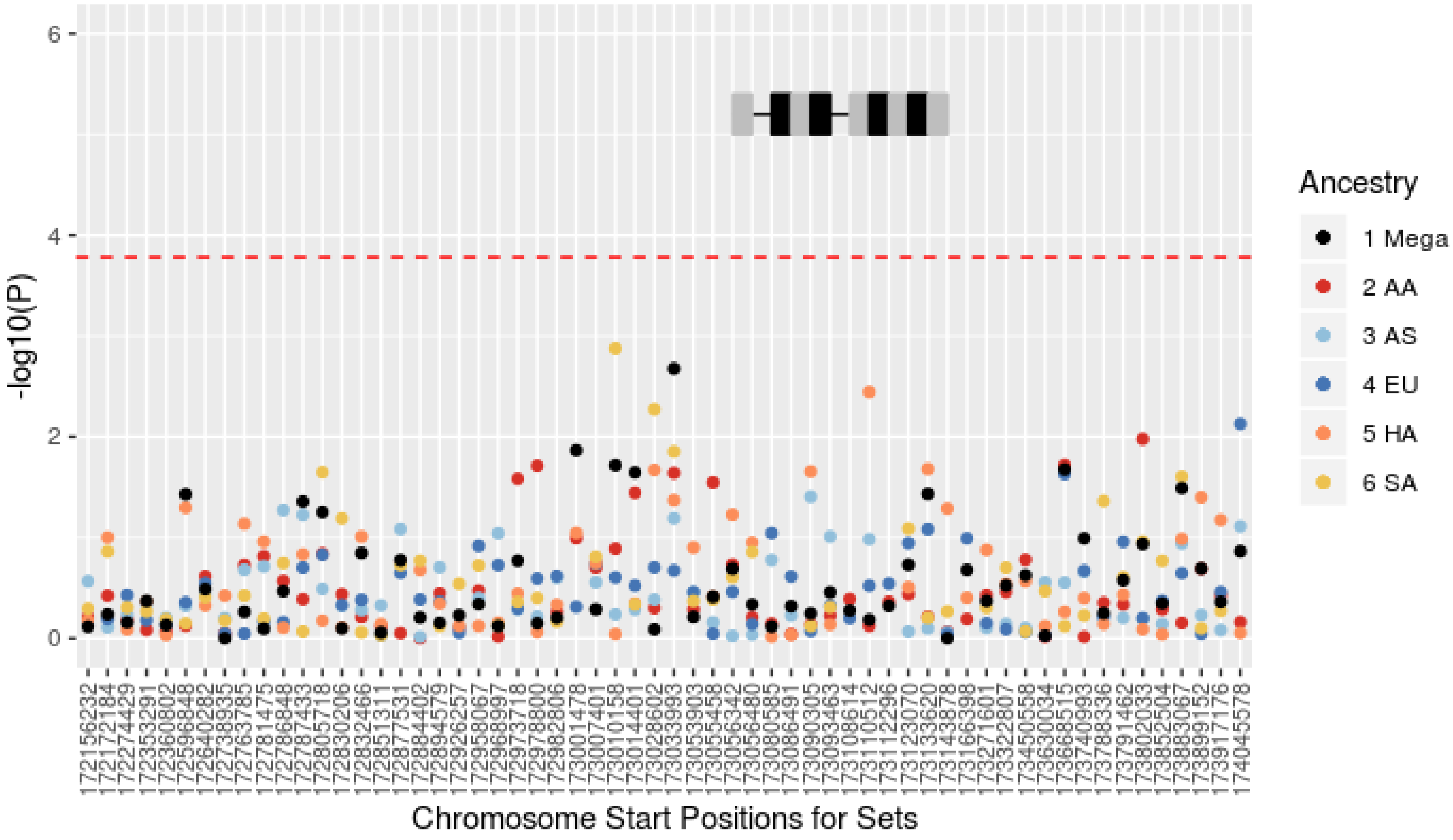
- Replace segments of sequence with abstract symbols
  - In gnomAD, introns have been abstracted to short segments



APOE. gnomAD. 2022 Feb 17

Abstraction

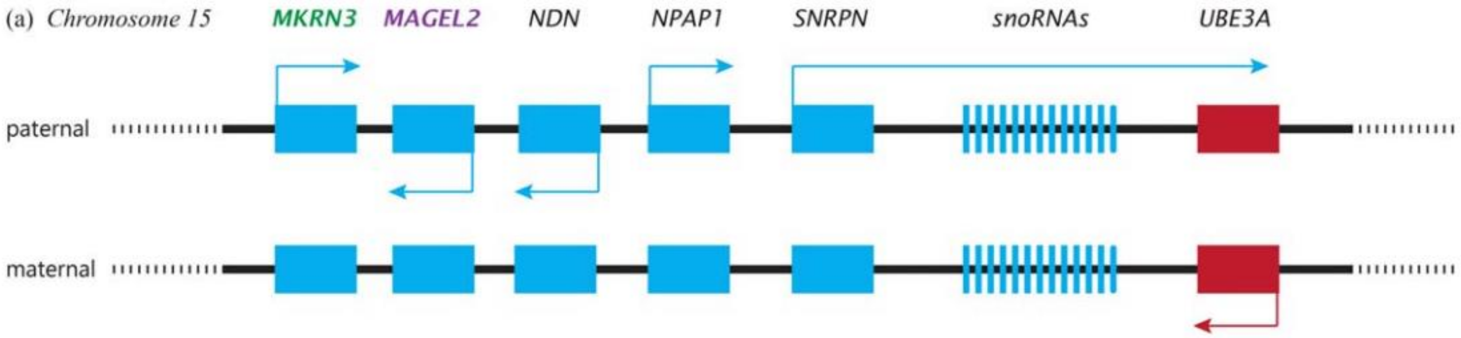
- Replace segments of sequence with abstract symbols



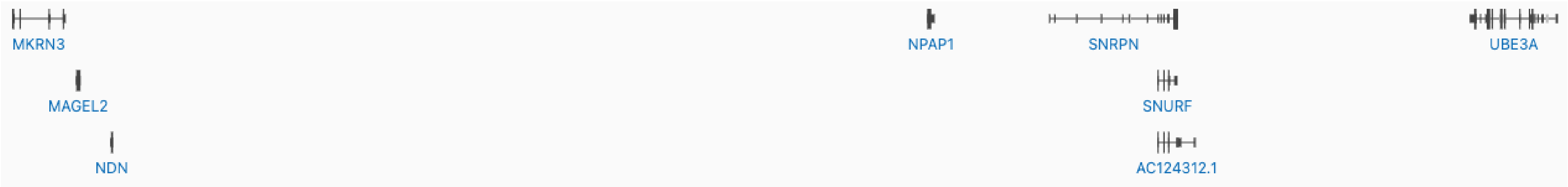
Russell *et al*. unpub.

Abstraction

- With increasing levels of abstraction, the **order and position** of features becomes **more salient** than the **relative lengths** of features.



Kotler & Haig. *Evol Anthropol.* 2018



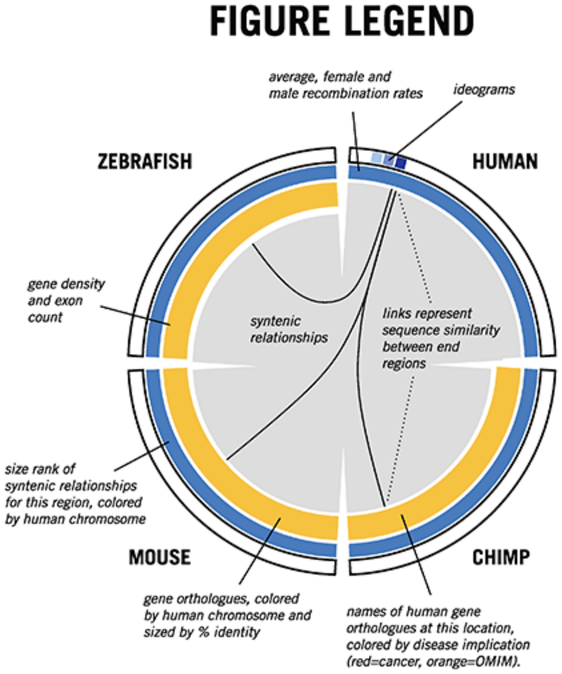
15:23565674–25630075. gnomAD. 2022 Feb 17

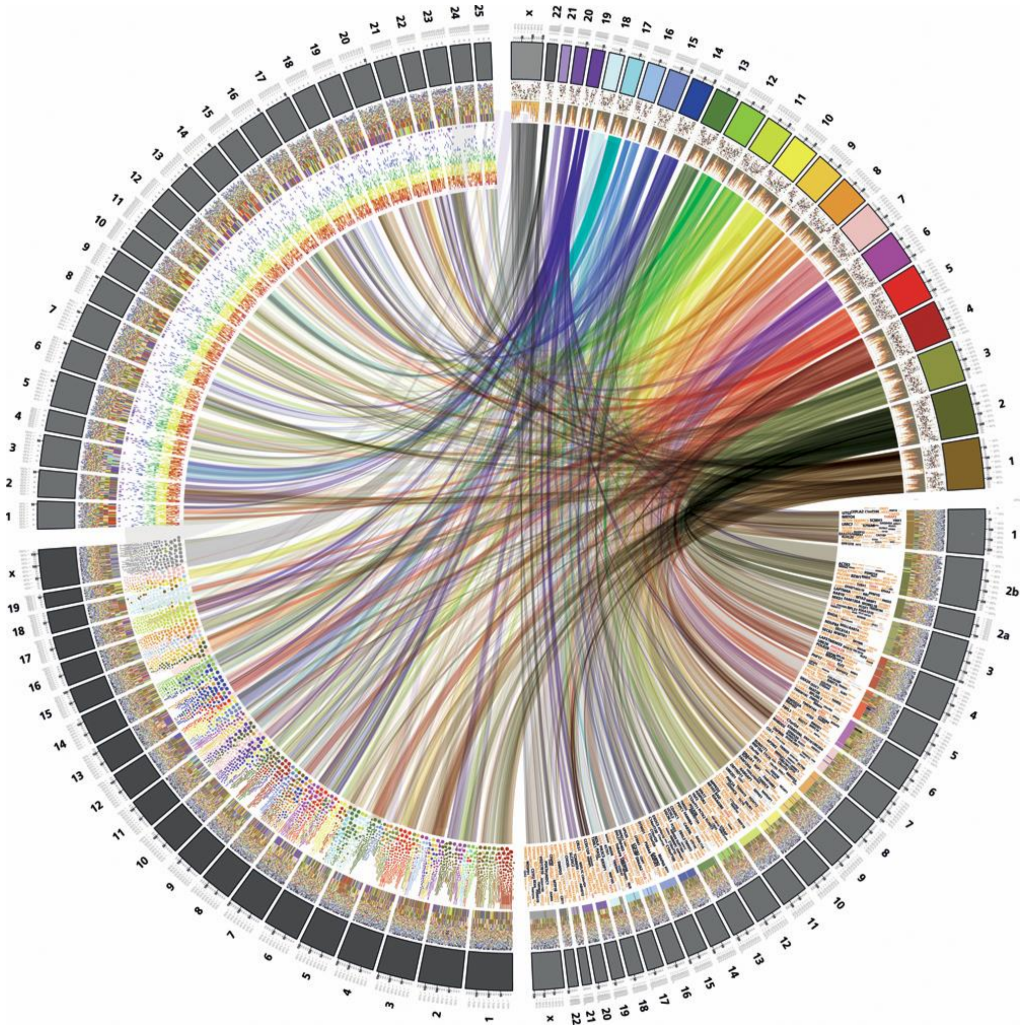
Arrangement (Comparing or Linking)

- Serial
- Parallel
- Orthogonal

Arrangement (Comparing or Linking)

- Serial Arrangement
  - End-to-end

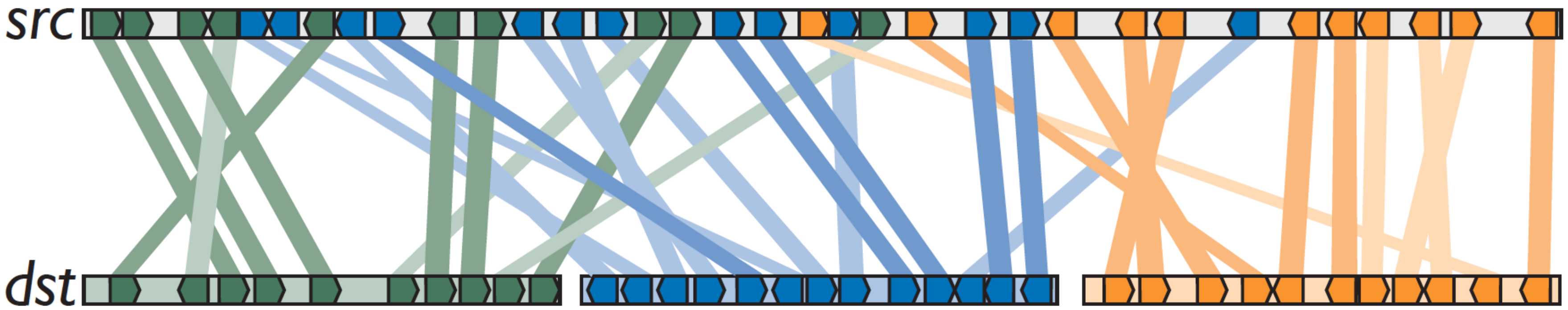




Krzywinski. http://circos.ca/guide/visual/img/circos-visualguide-med.png

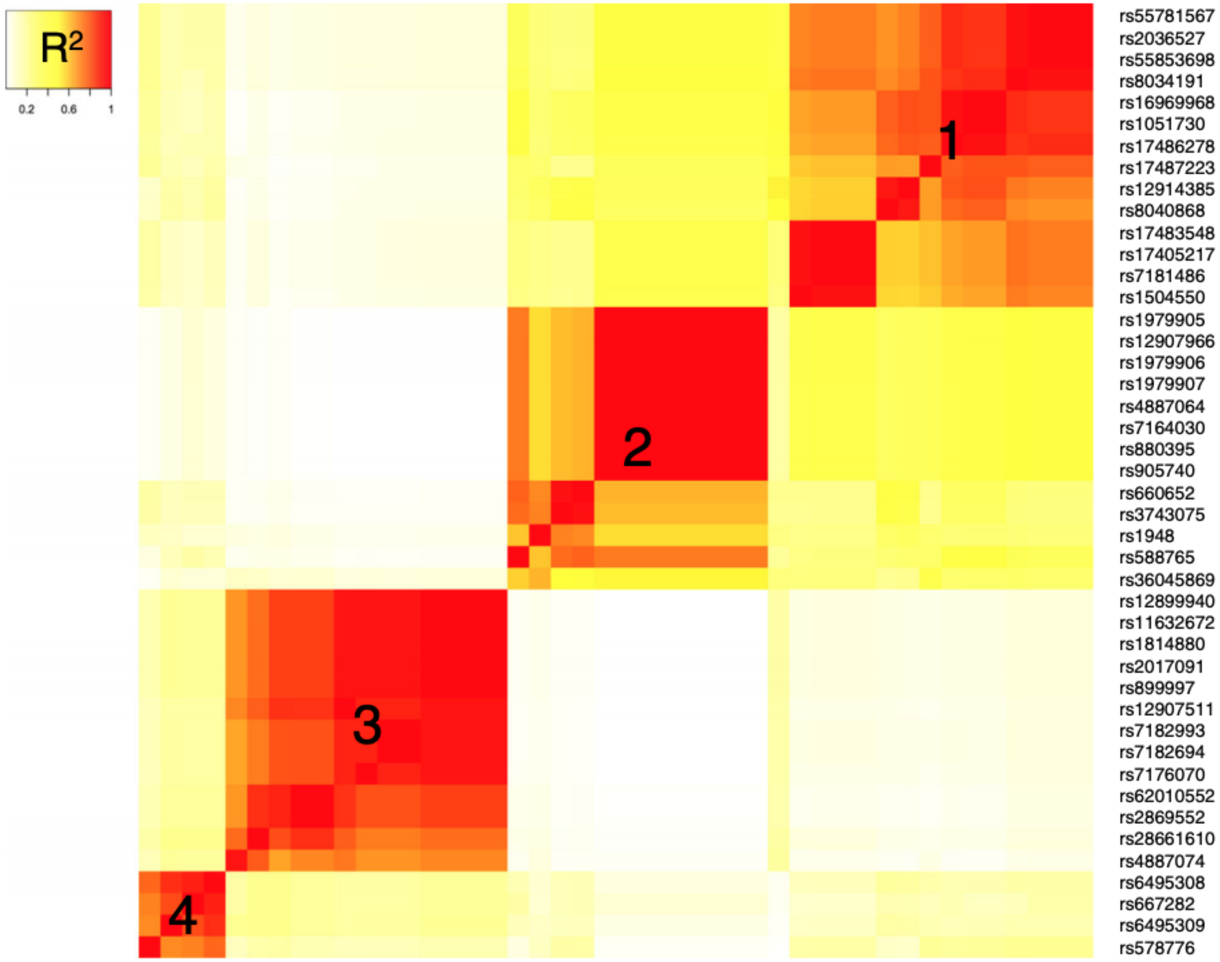
Arrangement (Comparing or Linking)

- Parallel Arrangement
  - Linear or circular



Arrangement (Comparing or Linking)

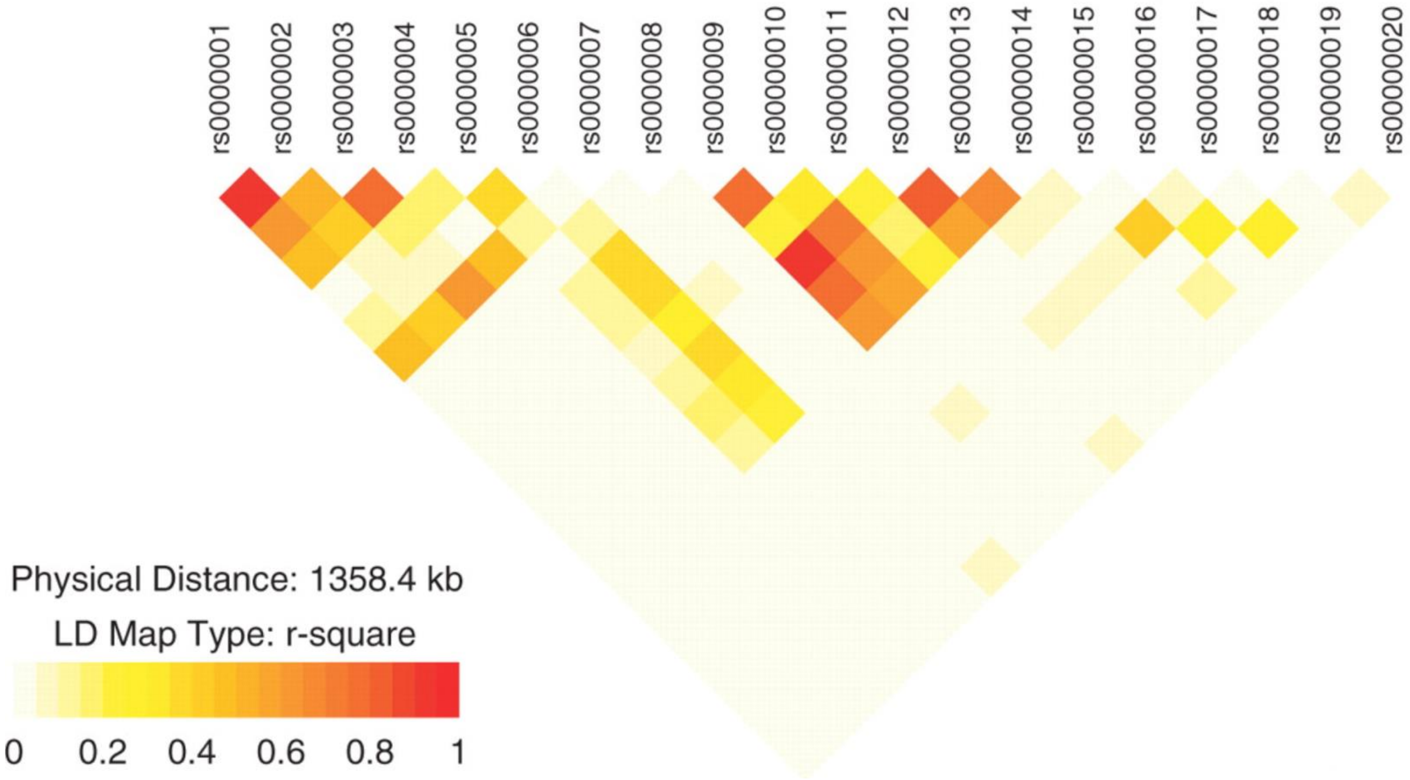
- Orthogonal Arrangement



Lee *et al*. *BMC Genomics*. 2018

Arrangement (Comparing or Linking)

- Orthogonal Arrangement (can often be rearranged to put the map along the diagonal)



Luna *et al*. *Bioinformatics*. 2007

Data or Feature Taxonomy

- Segmentation Type
- Length Type
- Attributes / Annotation

Data or Feature Taxonomy

- Segmentation Type
  - Sparse features
    - Genes, SNPs, etc.
  - Contiguous features
    - Copy number, coding/non-coding, etc.

Sparse Features

- Have a start and an end
  - Genes
  - Transcripts
  - Introns and Exons
  - SNPs
  - Regulatory Elements
  - Repeats

19-45409011-45427650. gnomAD. 2022 Feb 17

19-45408911–45424650. gnomAD. 2022 Feb 17

Contiguous Features

- Every point/segment in the genome has a “value”
  - Chromosome
  - Chromosome band
  - Copy number
  - Coding vs non-coding
  - Methylation
  - Acetylation
  - GC%
  - Sequencing coverage
  - Conservation

19-45408911–45424650. gnomAD. 2022 Feb 17

19-45408911–45424650. gnomAD. 2022 Feb 17

Data or Feature Taxonomy

- Length Type
  - Point features
    - 1 nucleotide
  - Segment features
    - > 1 nucleotide

- Both sparse and continuous features have lengths
  - Sequencing coverage is a point feature
  - (Useful) GC% is a segment feature

Data or Feature Taxonomy

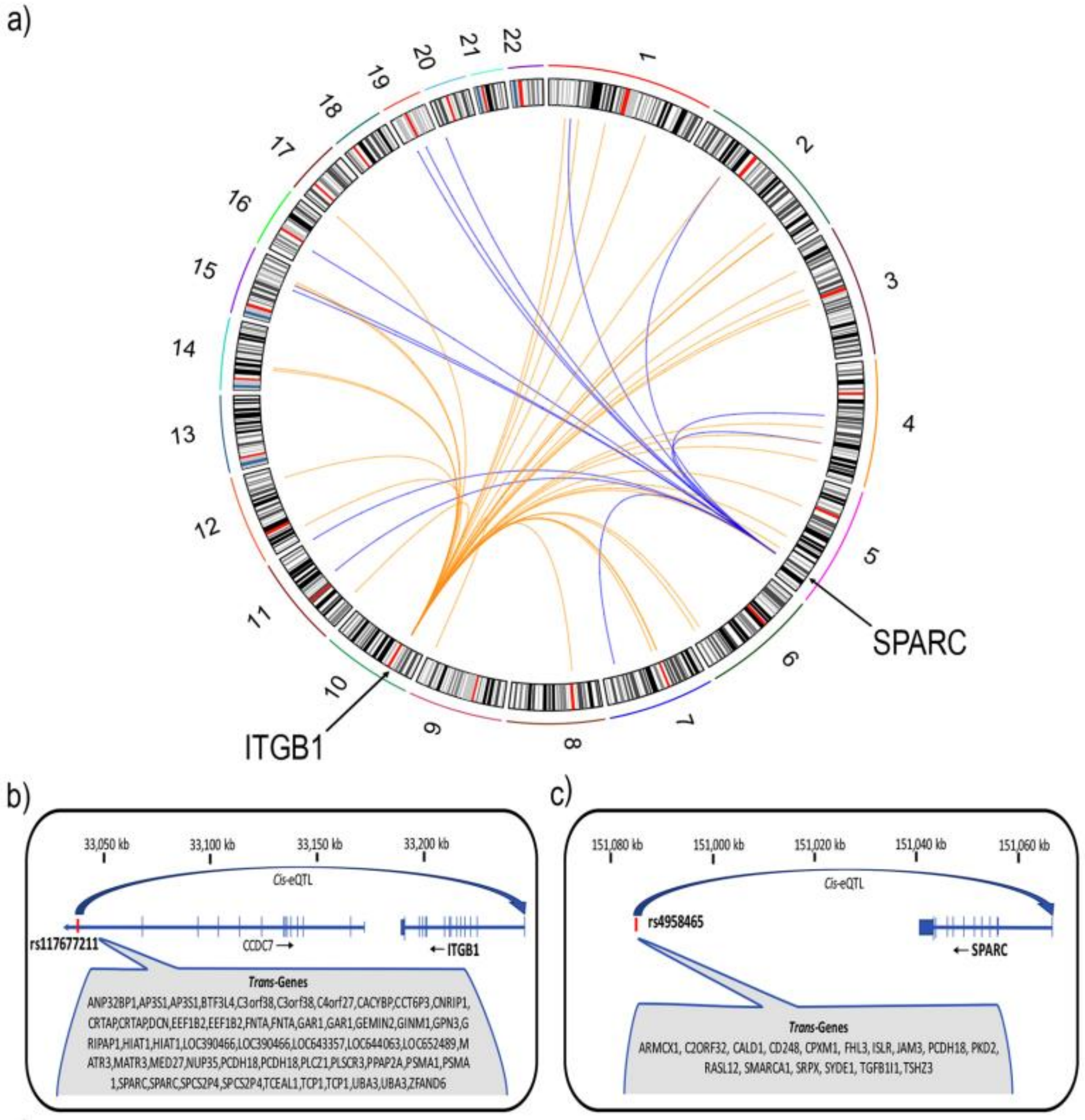
- Attributes
  - Number of attributes = 0, 1, or > 1
  - Values
    - Quantitative
      - Expression, copy number, LD, *p* values
    - Ordinal
      - Copy number
    - Categorical
      - Samesense, missense, nonsense

Data or Feature Taxonomy

- Interconnection
  - Links between features
    - eQTLs or other Regulatory Elements
    - Translocation
    - Co-location
    - Co-expression
    - Linkage disequilibrium
    - etc.

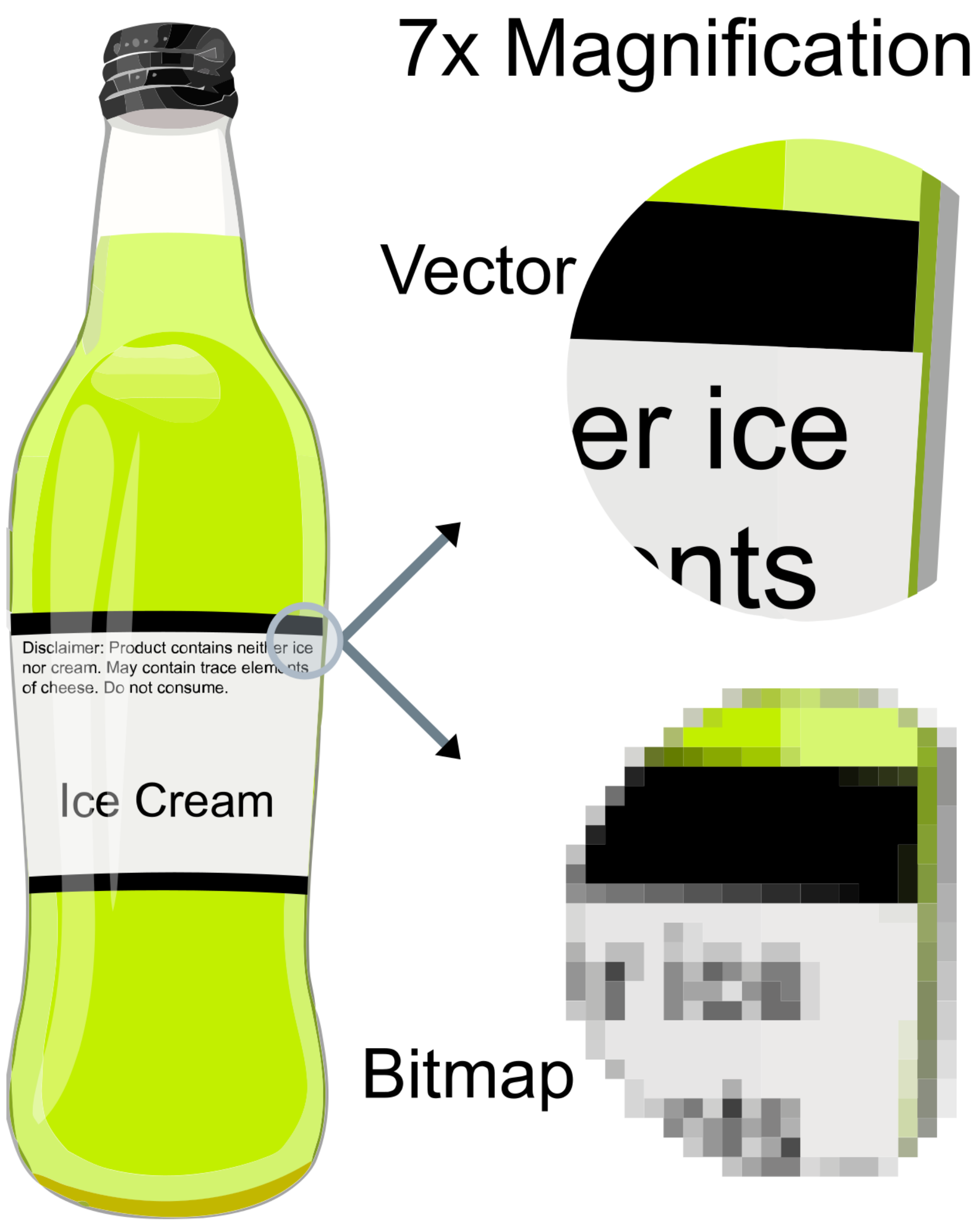
Visualizing Links

- These figures illustrate connections between eQTLs and the genes they are associated with
- The circos plot, (a), shows *trans*-regulatory links
- The linear plots, (b) and (c), show *cis*-regulatory links



Fung *et al*. *Sci Rep*. 2018

Vector vs Raster Graphics



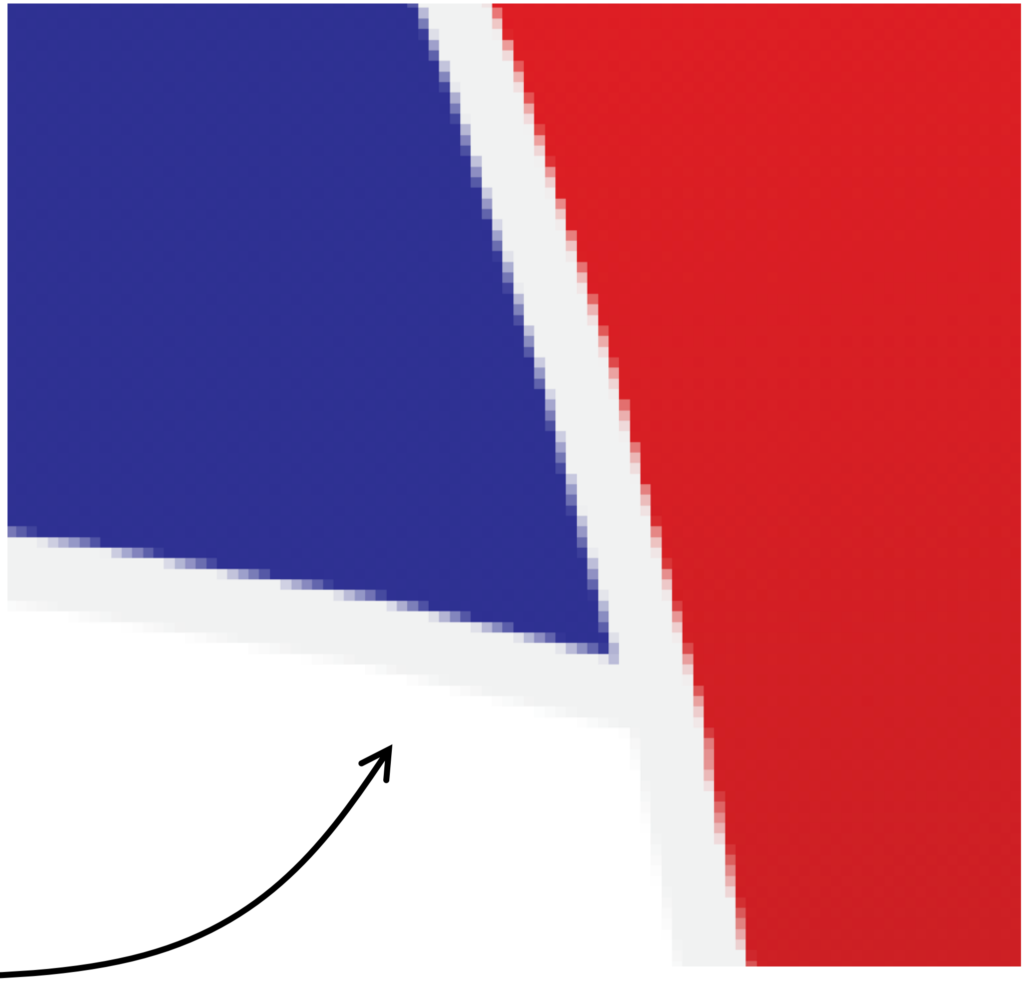
Wikipedia

Vector vs Raster

https://upload.wikimedia.org/wikipedia/commons/3/30/Vector-based_example.svg

If you go to the link and zoom in and in and in and in and in, it always remains sharp.

However, if you take a screenshot and zoom in and in and in, you see

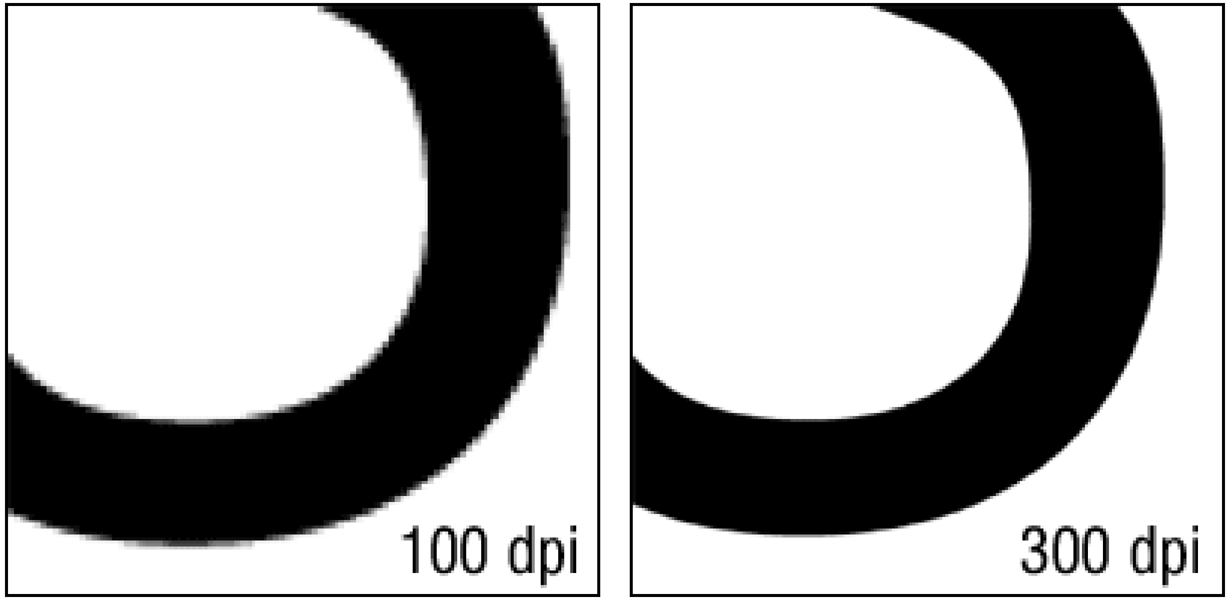


Vector vs Raster Graphics

- **Vector graphics** are **usually far preferable** for visualizations
- However, do not use vector graphics solutions for plots that have a ton of points
  - What are ”a ton of points”?
  - It depends on the processing capacity of the computer generating *or* (!!) displaying the image
  - Even if you have a high-powered machine that can create a vector graphics plot with 10,000,000 data points in it, no one on a low-power machine will ever be able to open it

Vector vs Raster Graphics

- If using raster graphics because of this limitation, turn up the resolution to at least 300 dpi
- But remember if the image gets enlarged the resolution drops
- So, an image at 300 dpi, if zoomed in 3× becomes 100 dpi



Vector vs Raster Graphics

- Journals will usual have standards for minimum dpi for raster images

File Formats and Suffixes

Vector Graphics

- **SVG** (scalable vector graphics)
- **EPS** (encapsulated postscript)
- **PDF** (portable document format)

Raster Graphics

- **PNG** (portable network graphics)
- **JPG** (joint photographic experts group)
    - **JPG** format often uses additional compression that further reduces quality
- **GIF** (graphics interchange format)
- **BMP** (bitmap)

Saving R Images

- To save images from R, you can use these commands:
  - ggsave()
    - Saves the last plot and guesses the format from the filename suffix
    - ggsave("myplot.pdf", width = 8, height = 8, units = "cm",
      dpi = 300)
  - pdf(), png(), svglite(), jpeg(), etc.
    - Opens a stream, saves plotted code to it.
    - It is necessary to close the “device” when done plotting.
    - png("myplot.png", w = 8, h = 8, u = "cm", r = 300)
      plot(…)
      dev.off()

Genomic Visualization Part 2

HUGEN 2073

Genomic Data Visualization & Integration

Slides adapted from Ryan Minster with permission

Learning Objectives

By the end of the session, students will be able to:

- Create Manhattan and QQ plots for genome-wide statistics
- Create regional Manhattan plots (LocusZoom plots)
- Use basic ggbio commands to create chromosome ideograms and plots of genes

Manhattan Plots

Principles

- x axis: genomic position (chr:bp)
- y axis: $-\log_{10} p$
- Point colors
  - Distinguish chromosomes
  - Distinguish important features
    - statistical significant “loci” (sometimes also point size)
    - known vs novel “loci”
- Reference lines
  - Genome-wide statistical significance (5×10^{-8})
  - Suggestive significance (often 10^{-5} or 10^{-6})

Code

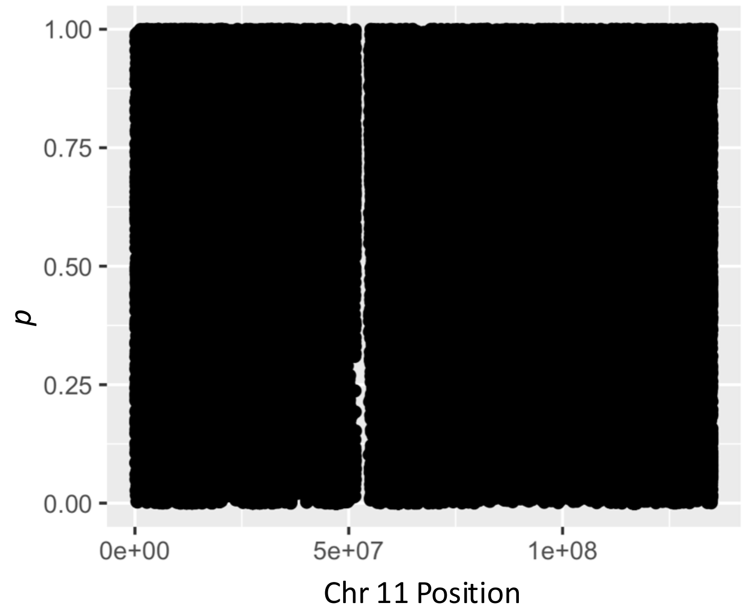
- Examples: **genome.sph.umich.edu/wiki/Code_Sample:_Generating_Manhattan_Plots_in_R**
- Examples: **https://r-graph-gallery.com/101_Manhattan_plot.html**

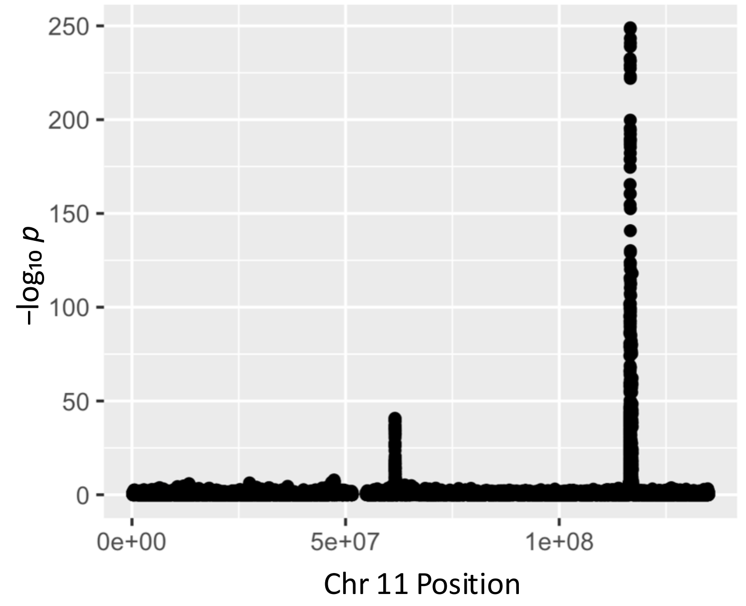
- Data: chromosome and position, *p* values from GWAS, SNP ID

```
library(qqman)
manhattan(results, chr = "CHR", bp = "BP",
                snp = "SNP", p = "P" )
```

Why $-\log_{10} p$?

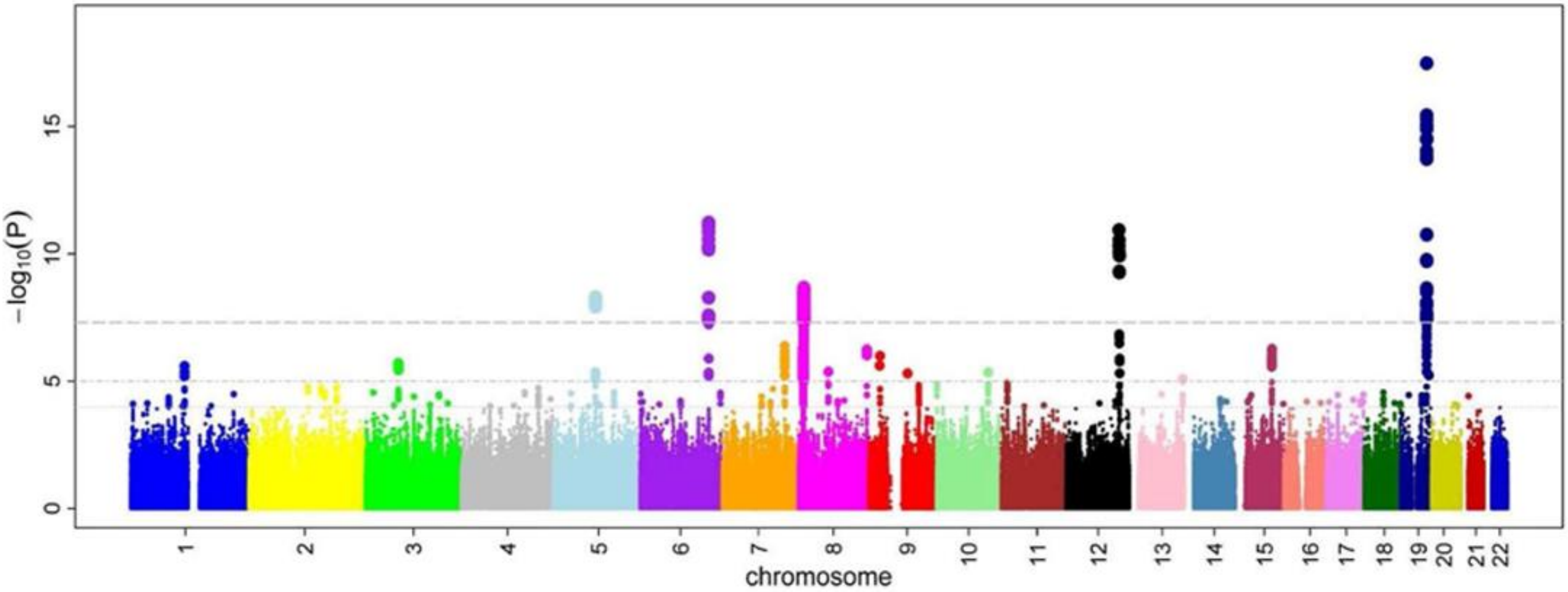
- *p* values are usually transformed
- Interesting *p* values (< 0.05) are compressed, and visually their differences are miniscule—whereas differences we don't care about like between $p = 0.5$ and $p = 0.7$ are large and visible
- $-\log_{10} p$ rescales *p* so that the interesting differences are visible
- The difference between 0.00000005 and 0.00001 is invisible but the difference between 7.3 and 5.0 is visible.





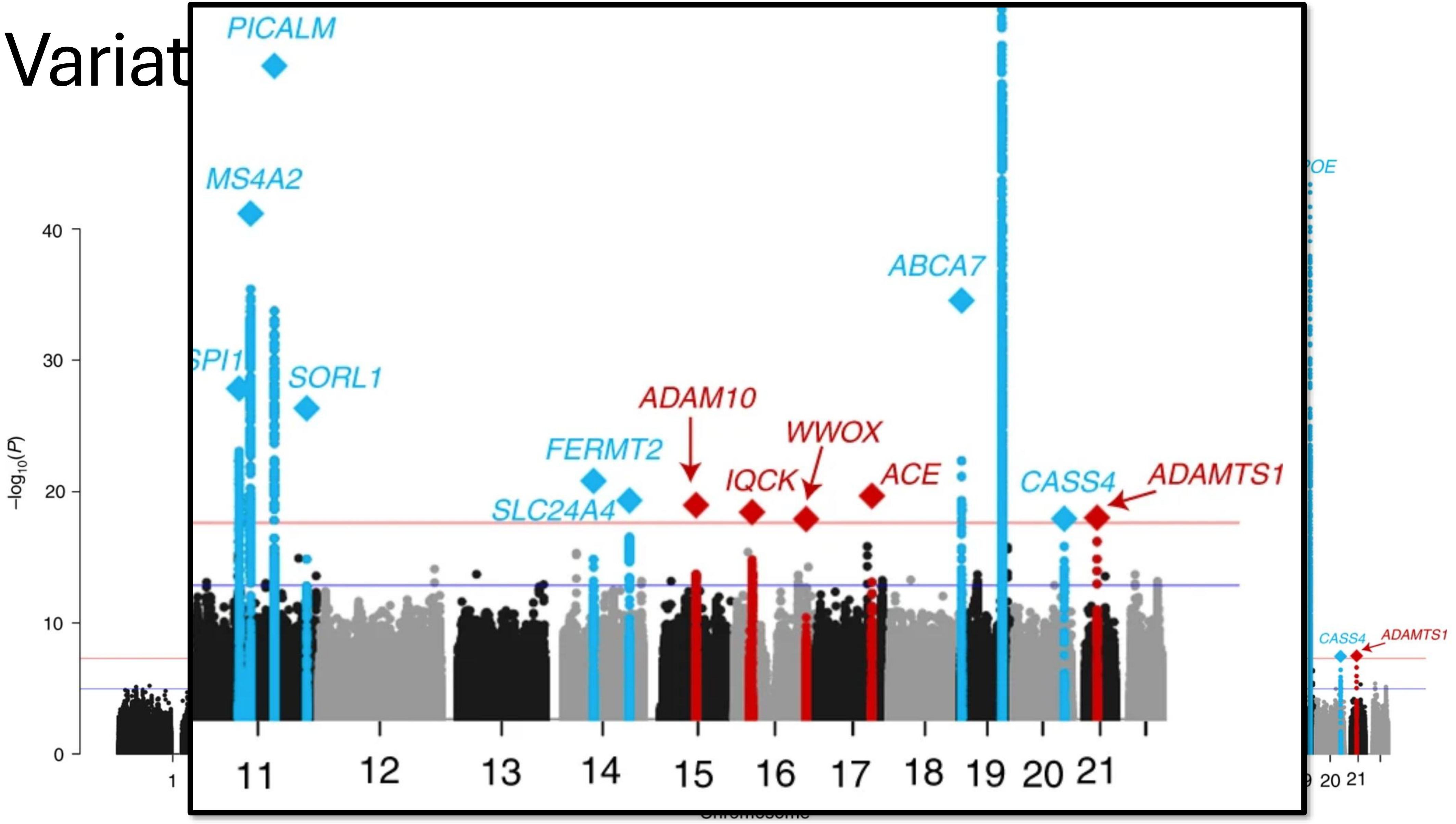
Data for plots from **Willer *et al.* *Nat Genet* 2013**

Variations



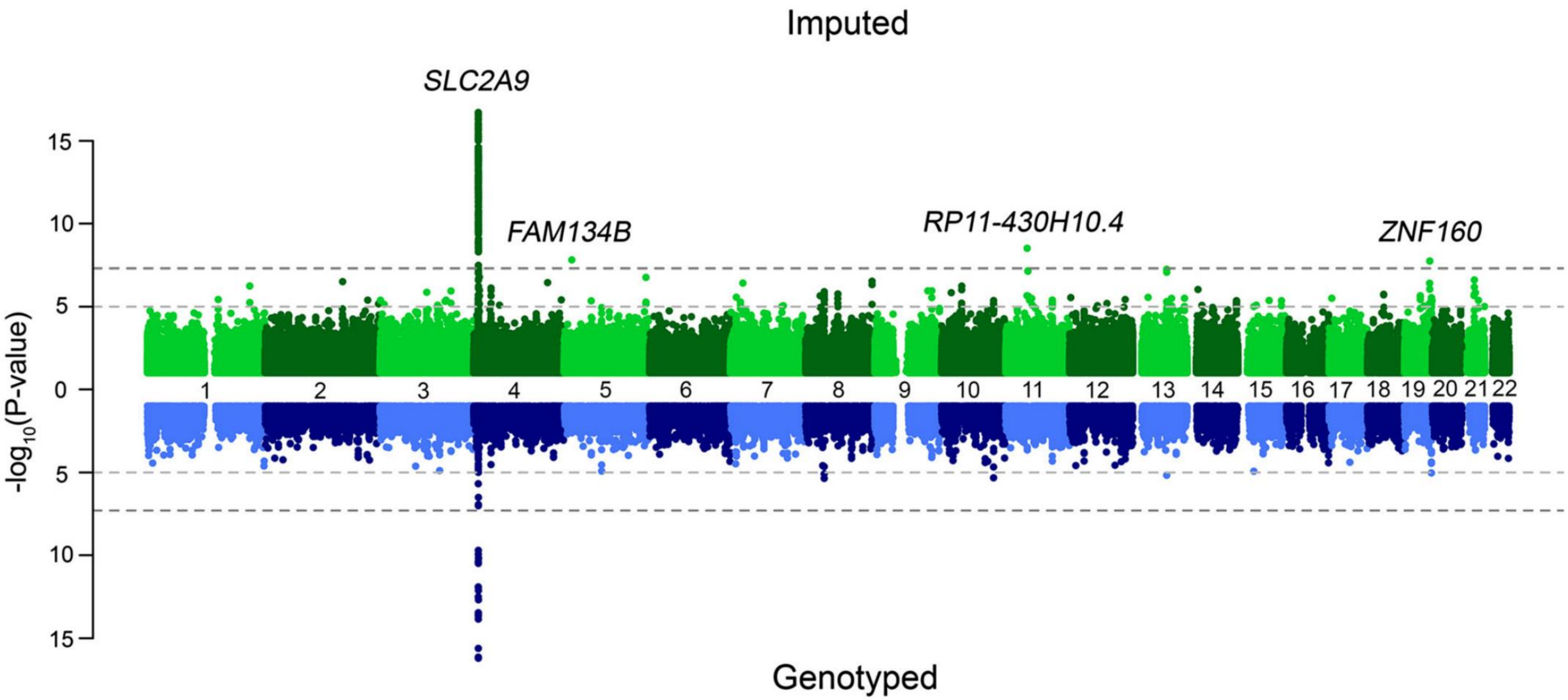
Ikram *et al*. *PLoS Genet.* 2010

Variat



Kunkle *et al*. *Nat. Genet.* 2019

Miami Plots

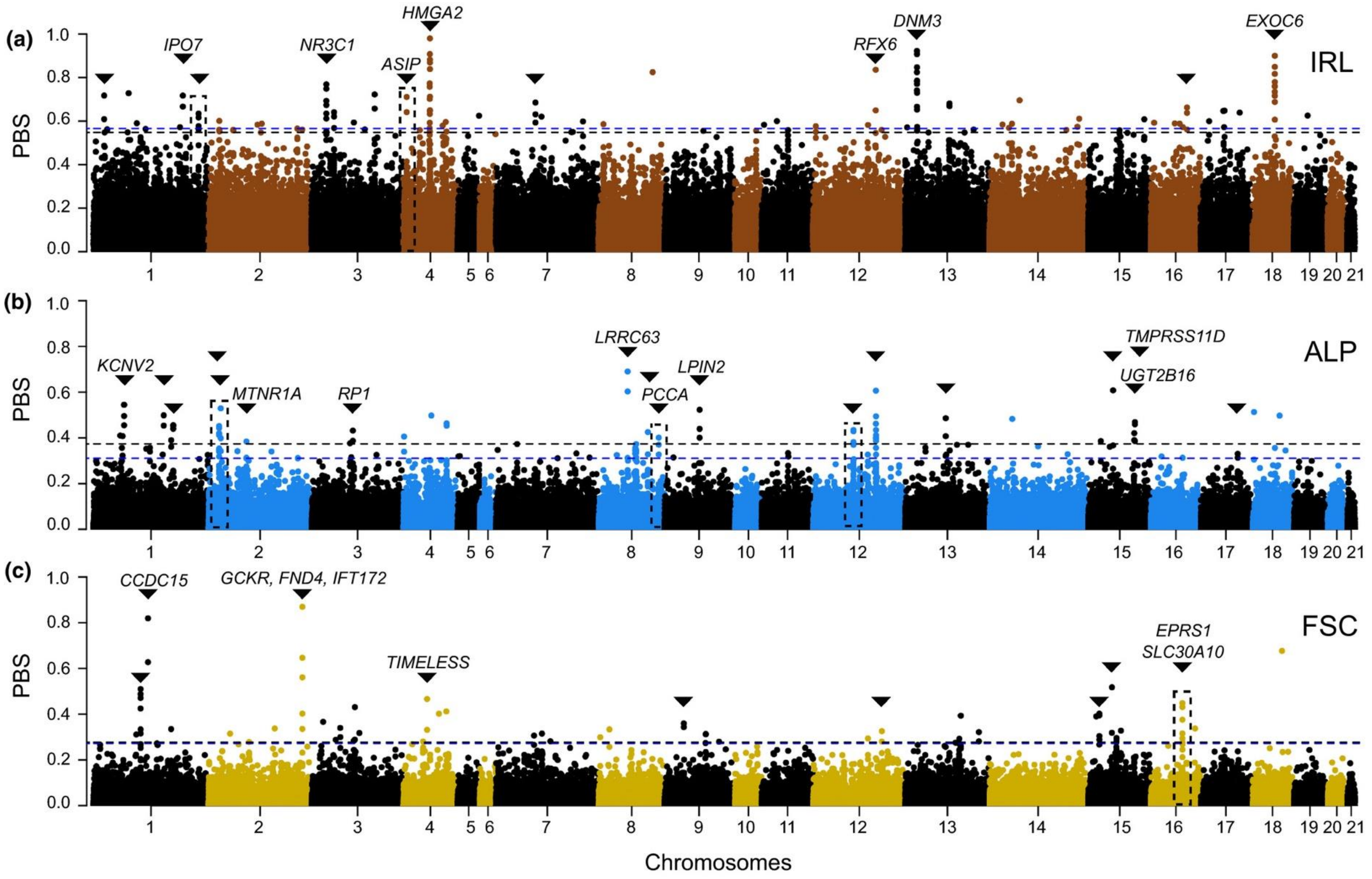


Nagy *et al. Genome Medicine* 2017

Beyond Association Test *p* Values

- Any point statistic can be used in a Manhattan-style plot
- Most common is *p* value ($-\log_{10} p$)
- You *could* plot t or F or χ^2 statistics — why not?
- Population genetics statistics (F_{ST}, r^2, D', PBS, nS_L)

Natural Selection



Giska *et al. Molec Ecol.* 2022

Shortcuts

- Reduce quantity of points to be plotted
  - Plot only points $-\log_{10} p > 2$
    - 98.998% of points in a GWAS will be below this threshold
    - Takes 1.5 million points down to 15,028
  - Plot only random subset of points $-\log_{10} p < 5$
  - Plot subset of chromosomes

QQ Plots

Principles

- x axis: expected $-\log_{10} p$
- y axis: observed $-\log_{10} p$
- Reference lines
  - $y = x$ (identity line)
- Confidence interval

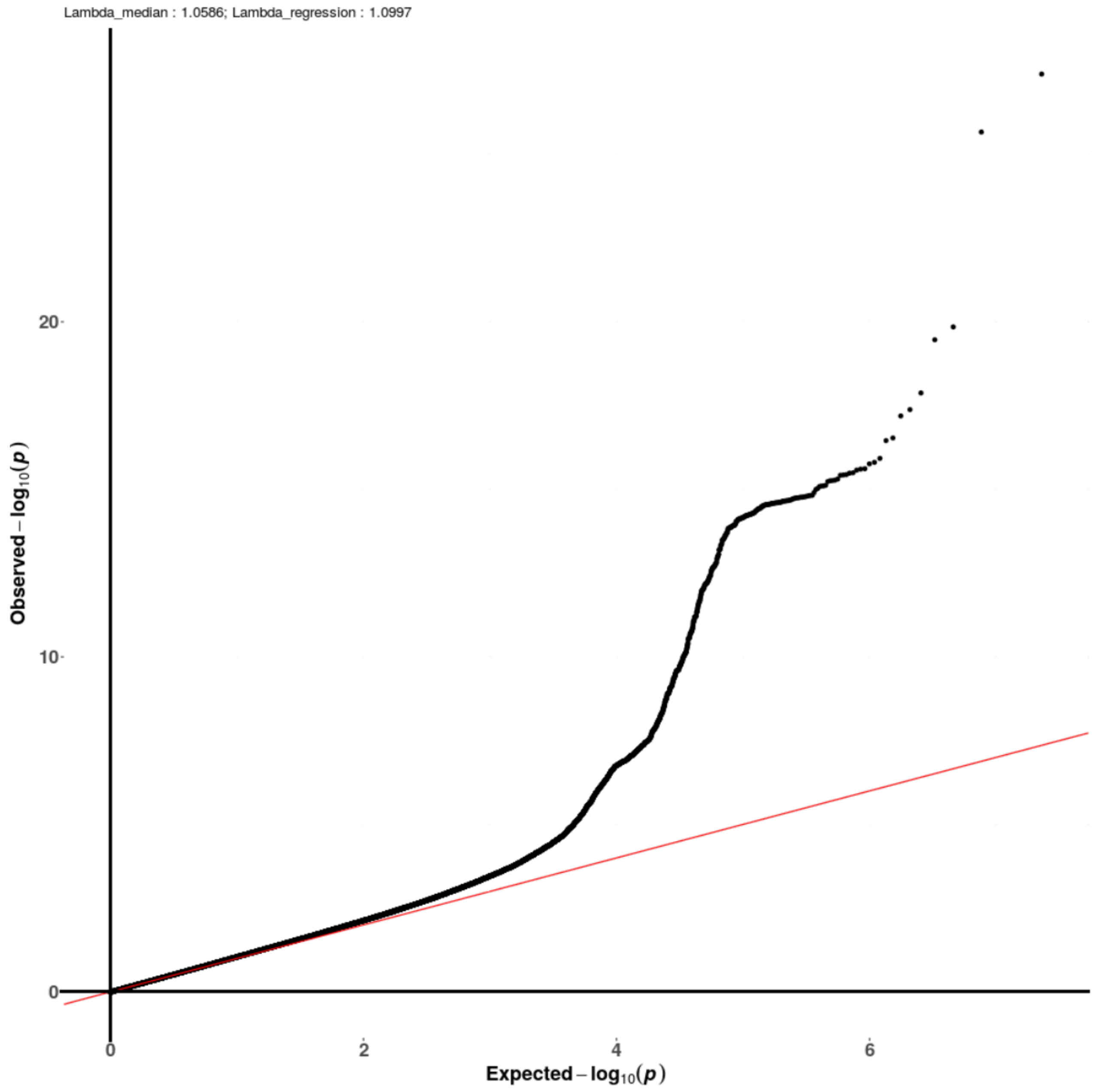
Code

- Examples: **genome.sph.umich.edu/wiki/Code_Sample:_Generating_QQ_Plots_in_R**
- Examples: **https://r-graph-gallery.com/101_Manhattan_plot.html**

- Data: *p* values from GWAS

```
library(qqman)
qq(results$p)
```

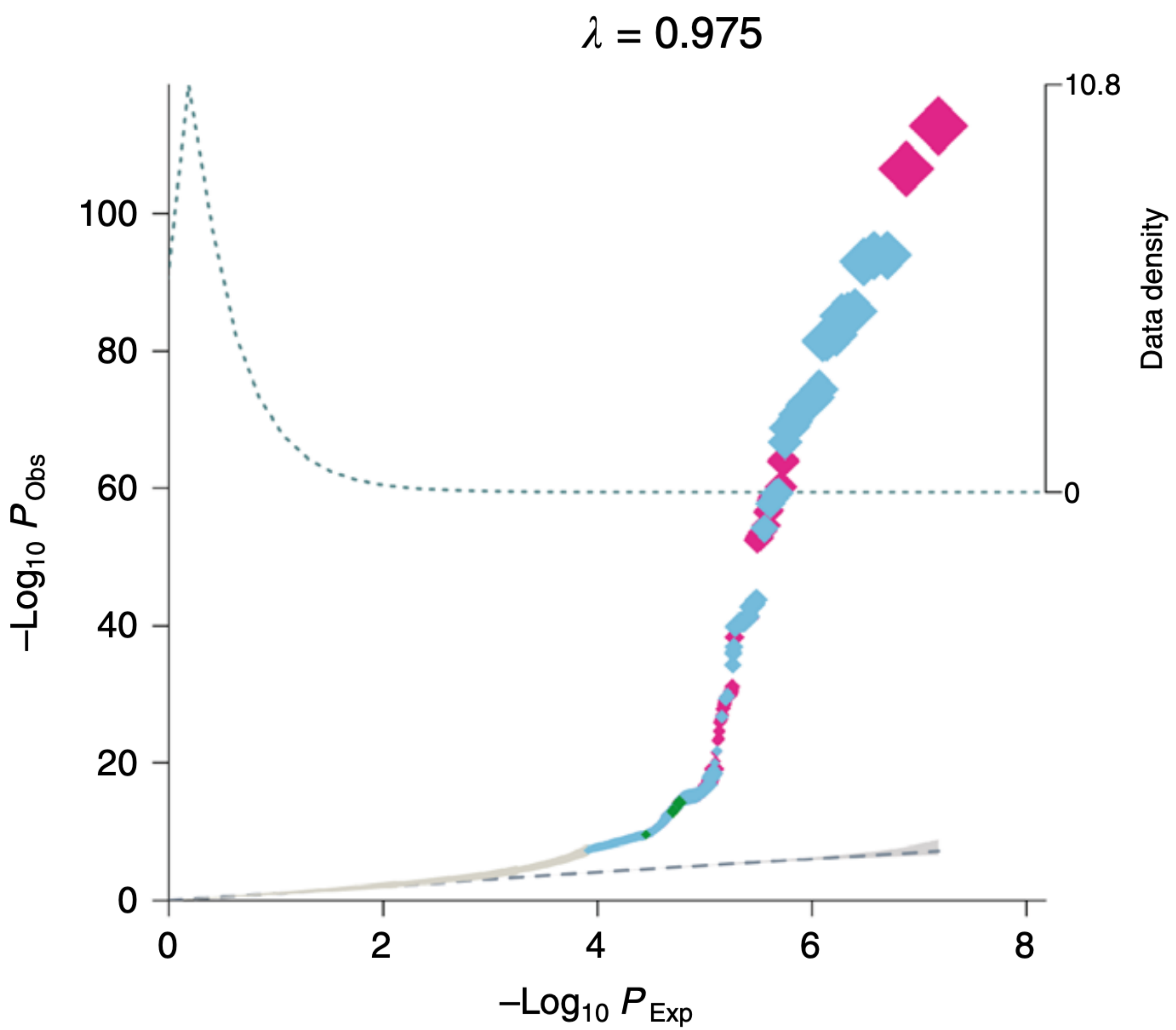
Variations



Kunkle *et al*. *Nat. Genet.* 2019

Variations

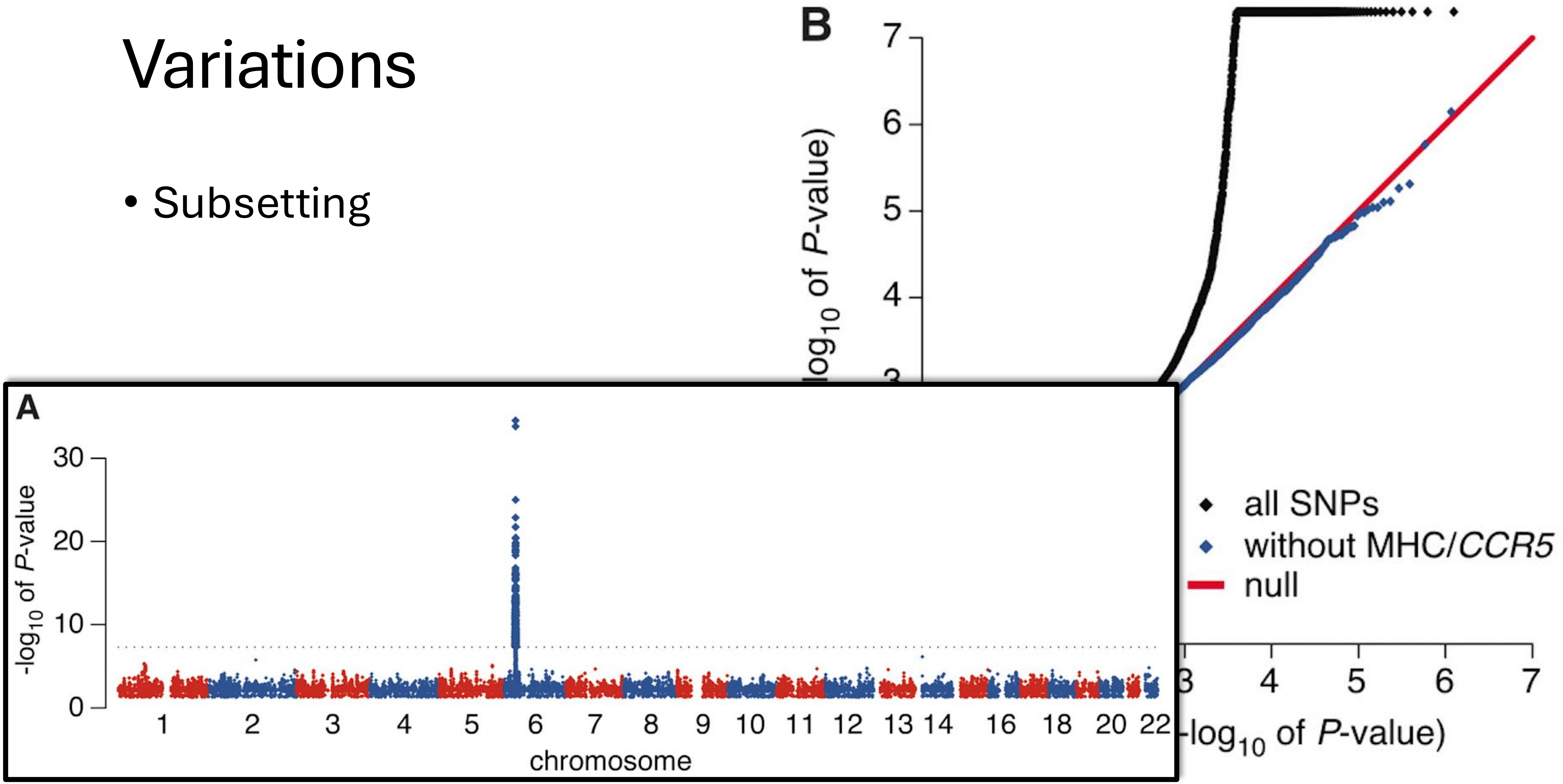
- Highlighting with size and color



Bonàs-Guarch *et al. Nat. Commun.* 2018

Variations

- Subsetting



International HIV Controllers Study; Pereyra *et al*. *Science* 2010

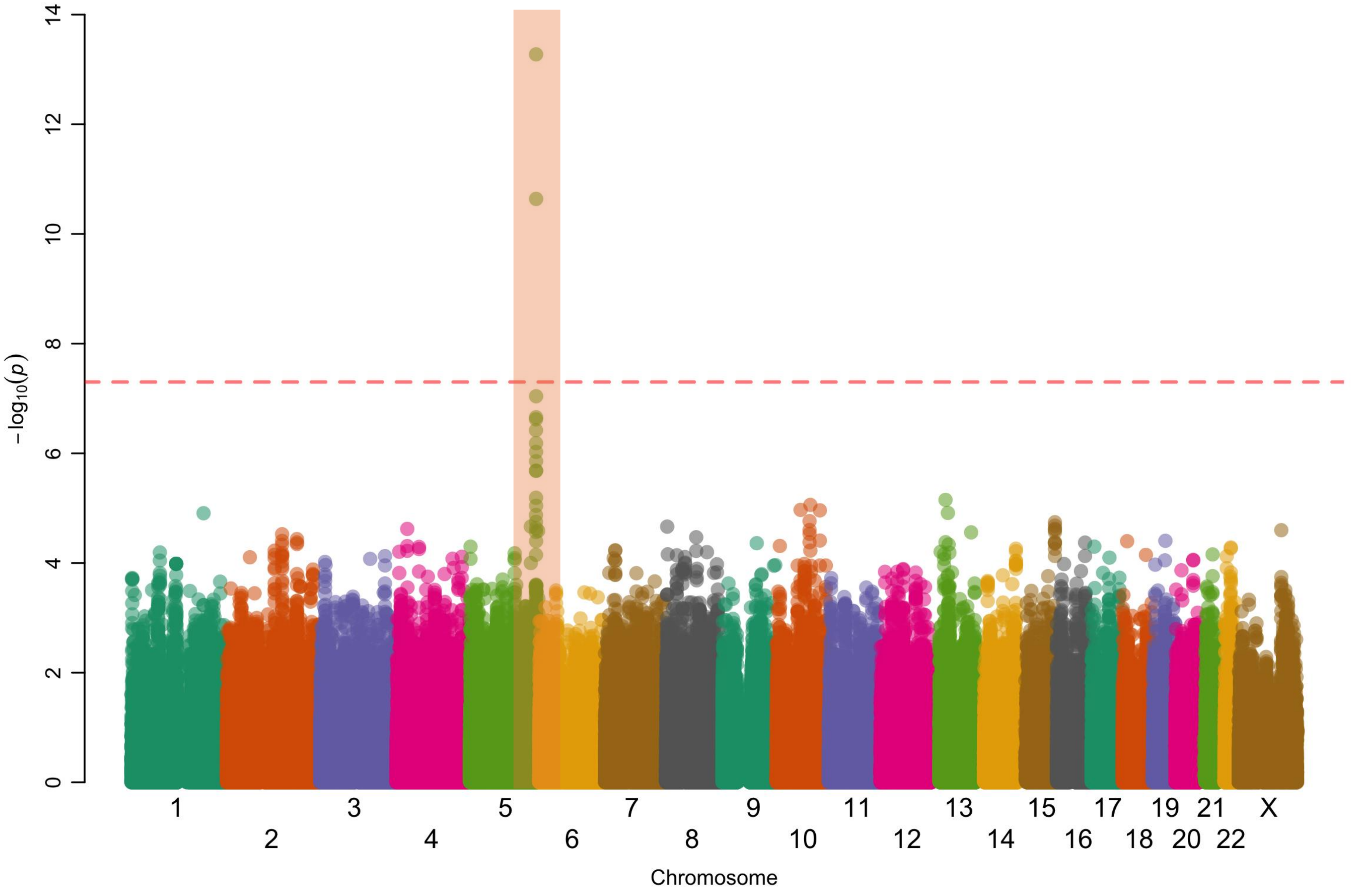
Vector vs Raster Graphics

- Vector graphics are usually preferable for visualizations
- Generally, do not use vector graphics solutions for QQ plots.
  - (PDF, SVG, EPS)
- Use raster graphics
  - (PNG, JPG, GIF)

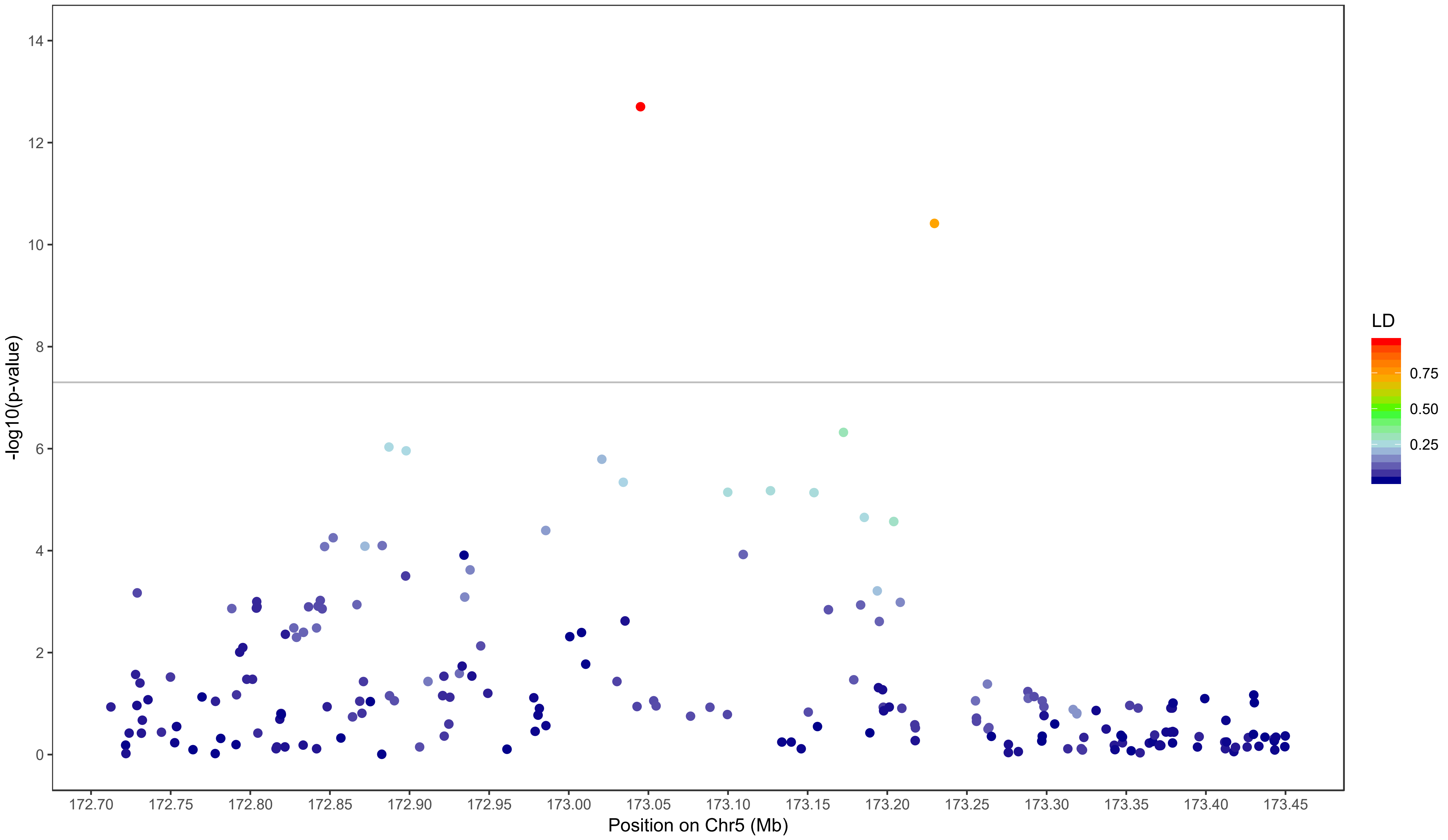
Regional Manhattan Plots

Regional Manhattan Plots

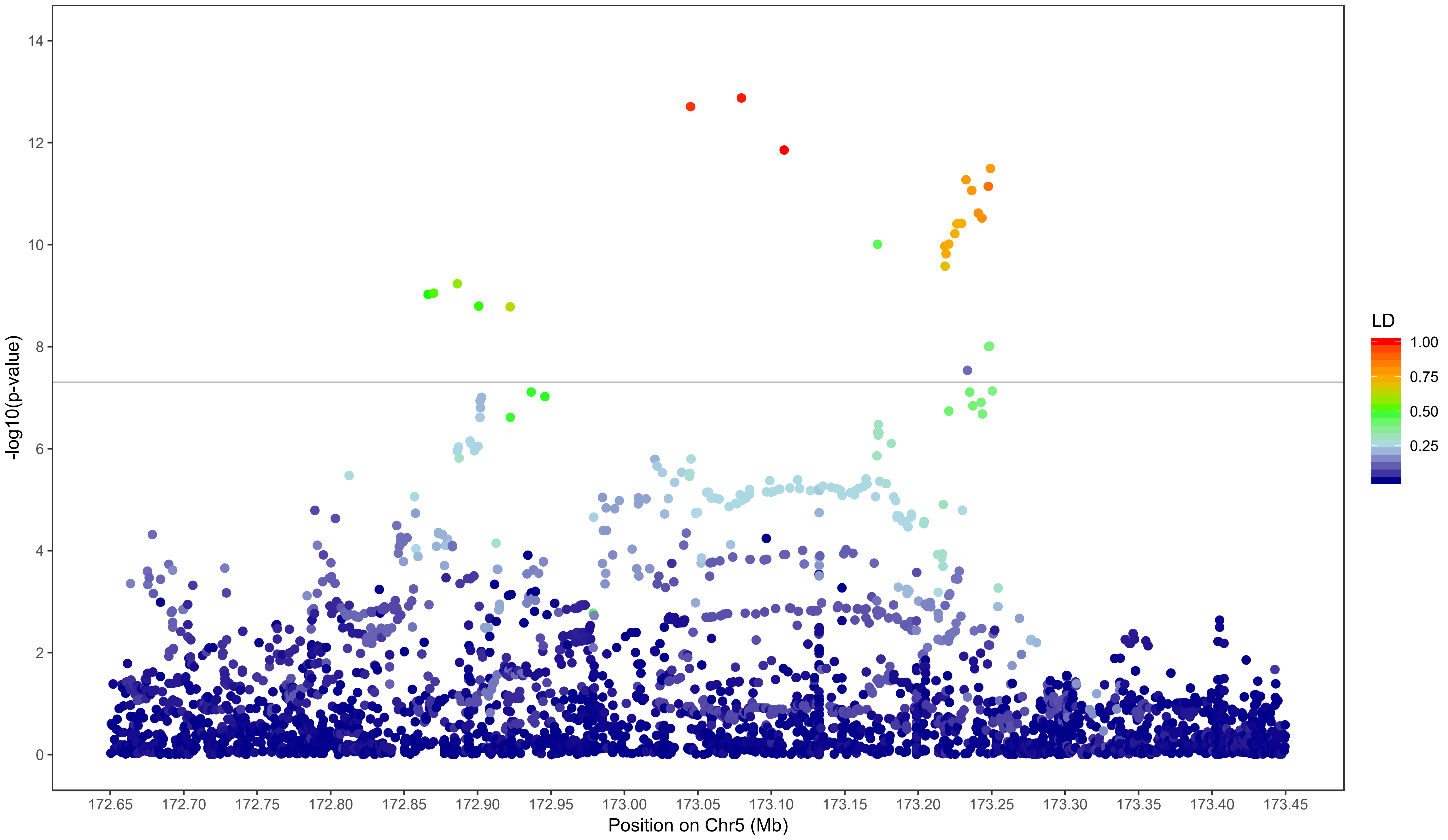
- Visualize small segment of genome



14
12
10
8
6
4
2
0
−log10(p)
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
X
Chromosome



-log10(p-value)
14
12
10
8
6
4
2
0
172.70
172.75
172.80
172.85
172.90
172.95
173.00
173.05
173.10
173.15
173.20
173.25
173.30
173.35
173.40
173.45
Position on Chr5 (Mb)
LD
0.75
0.50
0.25



-log10(p-value)
Position on Chr5 (Mb)
LD
1.00
0.75
0.50
0.25
0
2
4
6
8
10
12
14
172.65
172.70
172.75
172.80
172.85
172.90
172.95
173.00
173.05
173.10
173.15
173.20
173.25
173.30
173.35
173.40
173.45

Principles

- x axis: genomic position (chr:bp)
- y axis: $-\log_{10} p$
- Point colors
  - LD (usually r^2) between a “peak” SNP and the others
- Reference lines
  - Genome-wide statistical significance (5×10^{-8})
  - Suggestive significance (often 10^{-5} or 10^{-6})
- Often an overlay of recombination rate
  (to highlight recombination hotspots)
- Often underlaid with genome map include gene diagrams

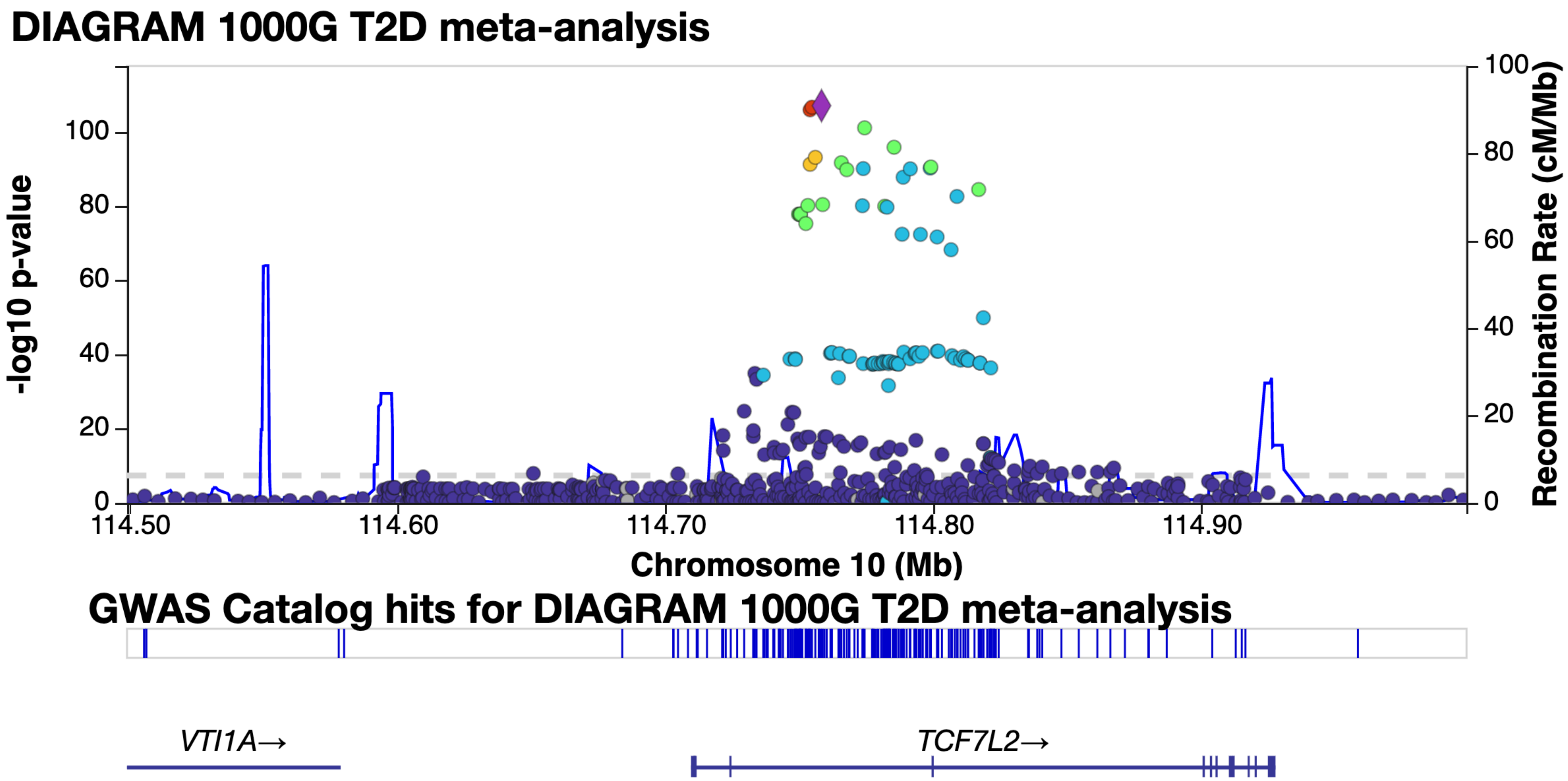


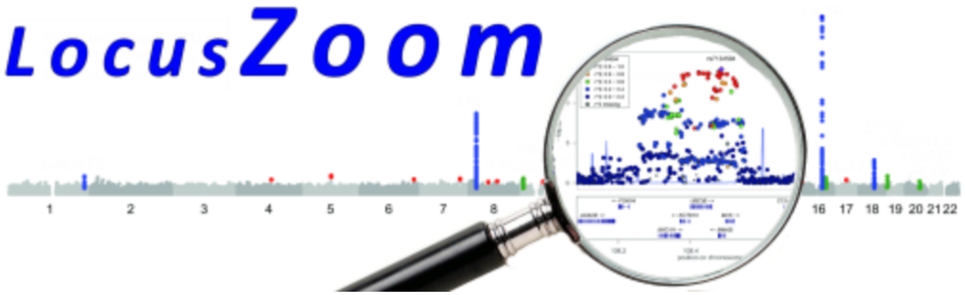
DIAGRAM 1000G T2D meta-analysis
-log10 p-value
0
20
40
60
80
100
Recombination Rate (cM/Mb)
0
20
40
60
80
100
114.50
114.60
114.70
114.80
114.90
Chromosome 10 (Mb)
GWAS Catalog hits for DIAGRAM 1000G T2D meta-analysis
VTI1A→
TCF7L2→

LocusZoom

- (So you don't have to code it yourself)
- LocusZoom

  locuszoom.org

  Pruim *et al. Bioinformatics* 2010



LocusZoom is a suite of tools to provide fast visualization of GWAS results for research and publication.

Original LocusZoom (R/Python) is ideal for batch generation of static plots.

LocusZoom.js (JavaScript) aims to make LocusZoom plots interactive and scriptable.

Interactive Plots with LocusZoom.js

MY.LOCUSZOOM.ORG
UPLOAD, ANALYZE, AND SHARE

LOCALZOOM
EXPLORE WITHOUT UPLOADING

Legacy Services (not actively maintained)

SINGLE PLOT
YOUR DATA - ORIGINAL LOCUSZOOM

BATCH PLOT WITH HITSPEC
YOUR DATA - ORIGINAL LOCUSZOOM

INTERACTIVE PLOT
PUBLISHED GWAS - LOCUSZOOM.JS

News

Planned Portaldev API downtime ~ November 13, 2020

We will be running operating system upgrades for the server that powers many of our data APIs (portaldev.sph.umich.edu). We have scheduled some downtime to install it next Friday Nov 13 @ 5:30PM (eastern US time). If you use interactive LocusZoom.js plots, you may notice either downtime or

Links

USER SURVEY

Interactive LocusZoom.js

LocusZoom

- Upload tab-delimited files with
  - Chromosome
  - Position
  - Reference allele
  - Alternate allele
  - p value or $-\log_{10} p$
- Will highlight regions above a threshold for LocusZoom plots

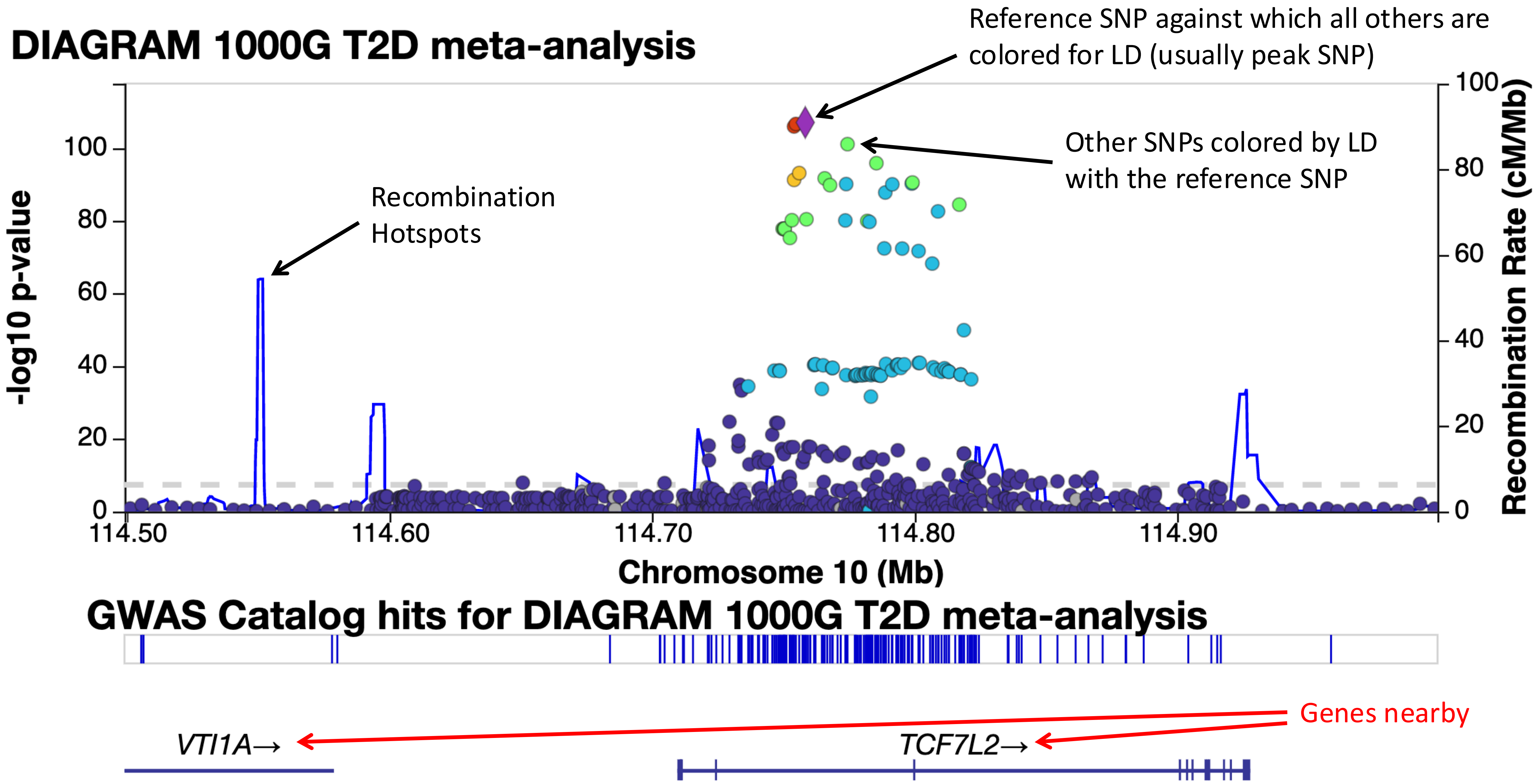


DIAGRAM 1000G T2D meta-analysis
Reference SNP against which all others are colored for LD (usually peak SNP)
Other SNPs colored by LD with the reference SNP
Recombination Hotspots
-log10 p-value
100
80
60
40
20
0
Recombination Rate (cM/Mb)
114.50
114.60
114.70
114.80
114.90
Chromosome 10 (Mb)
GWAS Catalog hits for DIAGRAM 1000G T2D meta-analysis
Genes nearby
VTI1A→
TCF7L2→

Standalone LocusZoom

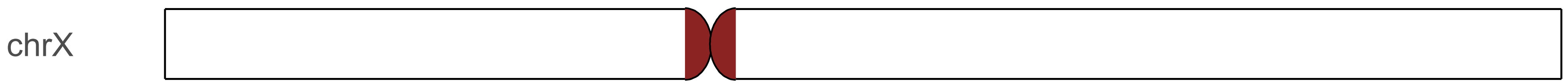
- There is a (no longer maintained) standalone version of LocusZoom
- Command line interface via Unix, built with Python and R

- **https://genome.sph.umich.edu/wiki/LocusZoom_Standalone**

ggbio

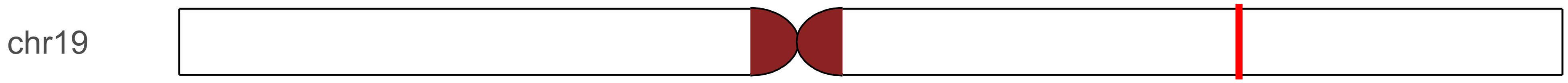
Plotting an Ideogram

```
library(ggbio)
plotIdeogram(genome = "hg38",
          subchr = "chrX")
```



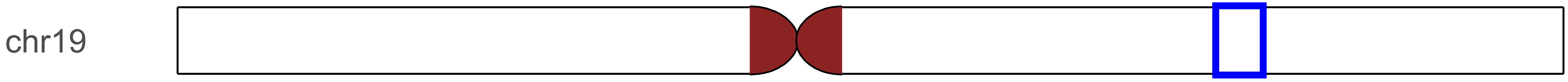
Plotting an Ideogram

```
library(ggbio)
plotIdeogram(genome = "hg38",
          subchr = "chr19",
          which = GRanges(seqnames = "chr19",
                    IRanges(44905791, 44909393))
```



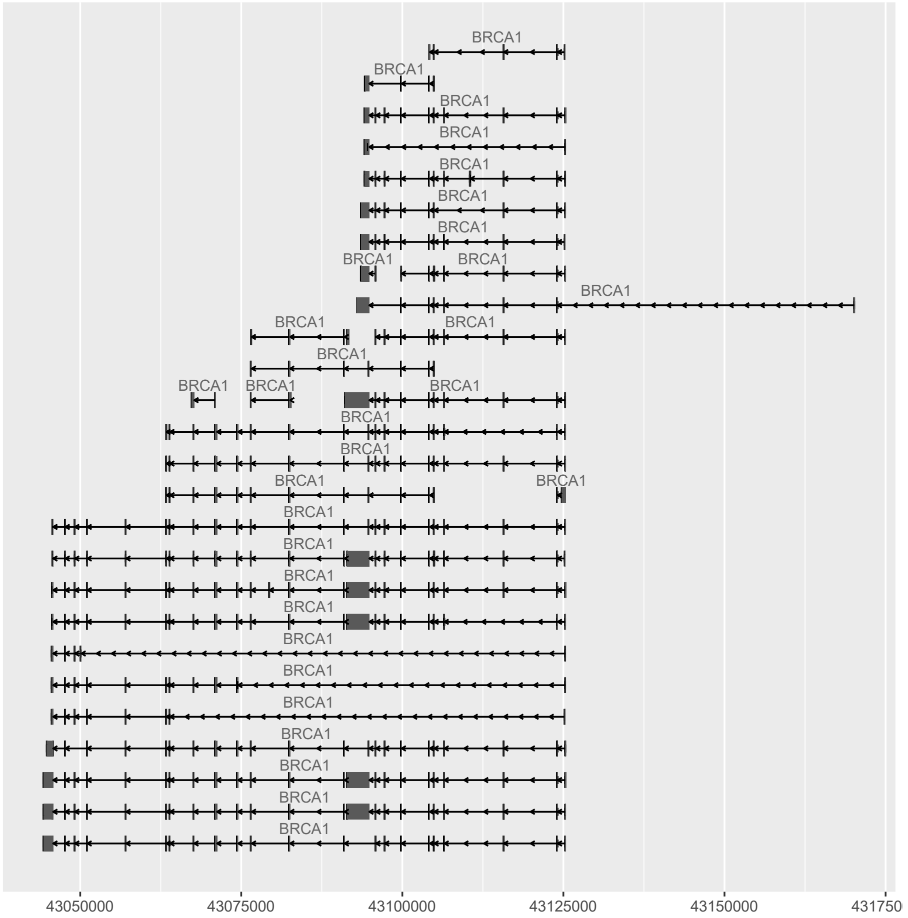
Plotting an Ideogram

```
library(ggbio)
plotIdeogram(genome = "hg38",
         subchr = "chr19",
         color = "blue", fill = "blue",
         which = GRanges(seqnames = "chr19",
                 IRanges(43905791, 45909393))
```



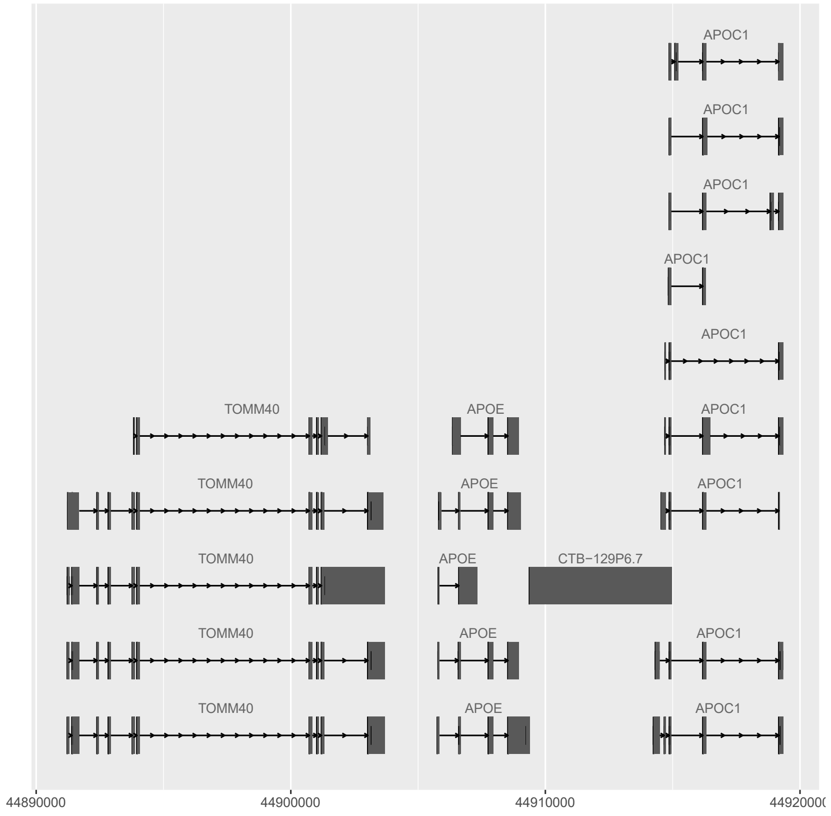
Plotting Genes

```
library(EnsDb.Hsapiens.v86)
autoplot(EnsDb.Hsapiens.v86,
        ~ symbol = "BRCA1",
        names.expr = "gene_name")
```



Plotting Regions

```
library(EnsDb.Hsapiens.v86)
gr <- GRanges(seqnames = 19, IRanges(44895791, 44919393),
           strand = "*")
autoplot(EnsDb.Hsapiens.v86,
       GRangesFilter(gr),
       names.expr = "gene_name")
```



Exercise

- Download summary statistics for a GWAS of your choice from https://www.ebi.ac.uk/gwas/downloads/summary-statistics
- Read the data into R
- Create Manhattan and Q-Q plots
  - Try two packages
    - qqman (https://cran.r-project.org/web/packages/qqman/vignettes/qqman.html)
    - GWASTools
      - Installation instructions: https://www.bioconductor.org/packages/release/bioc/html/GWASTools.html
      - Plotting functions:
      - https://rdrr.io/github/smgogarten/GWASTools/man/qqPlot.html
      - https://rdrr.io/bioc/GWASTools/man/manhattanPlot.html
  - Save them as appropriate file types
  - Can you speed up the plotting?
  - Can you customize the plots at all?